



Energy 4.0: AI-enabled digital transformation for sustainable power networks

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ABSTRACT

The considerable developments in green technology have changed several industries, including the energy business and power sector. These advances have transformed power generation and distribution. Artificial intelligence (AI), machine learning, computer vision, blockchain, and the internet of things (IoT) have advanced significantly in recent years. These advances have transformed the energy business and expanded its applications. In this concise yet comprehensive analysis, energy and power systems are examined from an eco-friendly perspective and emerging research trends are highlighted. The increasing use of renewable energy sources necessitates infrastructure restructuring, which presents the potential for grid development and AI usage for intermittent and distributed generation. The integration of AI, IoT, and blockchain technologies into energy and power systems offers opportunities to improve efficiency, sustainability, and dependability. Application-specific research is comprehensively examined to investigate the present state-of-the-art and limitations of various technologies. The focus in this study is the importance of implementing a complete policy and planning framework to support a significant and efficient shift toward advanced grid systems. The convergence of AI, information and communication technology, and energy systems offers unparalleled prospects affecting the trajectory of energy generation, distribution, and consumption in the future. This study makes a valuable contribution to the wider academic discussion on utilizing digital transformation to create sustainable and intelligent energy ecosystems by integrating insights acquired from several fields.

1. Introduction

Sustainability has become more important in organizations due to global concerns about greenhouse gas (GHG) emissions. The rapid growth of green technologies, such as the implementation of solar panels (Ahmed, Irshad, et al., 2023), the widespread acceptance of electric vehicles (Corradi, Sica, & Morone, 2023), and the deployment of wind turbines, has fundamentally transformed the energy sector (Charfed-dine & Umlai, 2023). United Nations Sustainable Development Goal 7 emphasizes the need for cheap and sustainable energy. The SDGs aim to give everyone cheap, dependable, and advanced energy services (Gao, Luo, Liu, & Fan, 2022). The term “contemporary” refers to energy and electrical innovation that involves creating a complete and efficient electrical grid. This holistic strategy revolutionizes energy and power systems from the beginning to the end.

When viewed separately, the energy sector contributes roughly 66% of global greenhouse gas (GHG) emissions, and within it, electricity generation contributes 40% of global GHGs (Iea, 2024b). The combined use of energy in power generation, manufacturing processes, and transportation accounts for a substantial 80% of the overall world carbon

dioxide (CO₂) emissions (Philibert, 2017). Academic, policy, and industrial sectors have emphasized green power sources in recent years, and the power sector has rapidly shifted to environmentally friendly energy sources, including wind power and solar systems. The Levelized Cost of Electricity has decreased due to solar module cost reductions; however, sustained growth requires more than power-producing sector changes (Philibert, 2017). Electricity management and consumption must change, and the sole use of clean energy for energy generation is insufficient to create a sustainable and dependable energy future that can effectively address global climate change. The problem needs a complete strategy. The suggested strategy should include enhanced energy storage materials and technologies, optimized and intelligent manufacturing processes, a sophisticated, resource-efficient, and transparent grid, and cautious energy resource management.

This is because the move to renewable energy sources (RESs), which is changing electricity generation, is not happening in a vacuum. The integration of RESs into the conventional grid presents management and maintenance problems. For example, a traditional grid cannot

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handle the power fluctuations associated with RESs, and therefore, how electricity is created, transported, allocated, used, and valued must be revamped. To this end, a thorough grid system reorganization is currently ongoing. Advanced sensors, bidirectional communication networks, and intelligent consumption devices have increased data-processing requirements. The innovation and integration of new technologies and regulations in energy production, management, and consumption are essential for building a sustainable and dependable power industry (Sodhro, Pirbhulal, & De Albuquerque, 2019). Embracing information technology (IT) 4.0 technologies is crucial for the global effort in shifting toward environmentally friendly, efficient, and intelligent solutions.

Digital technologies and frameworks like AI, IoT, Blockchain, and other data-centric techniques have great potential in this environment. For example, the authors of study (Danish & Senjyu, 2023) provide an unconventional AI-enabled model for contemporary resource legislation that reduce risks, encourage effective deployment, and forward society's goal of becoming zero emissions and carbon-free. The theory of Blockchain was investigated by Shahinzadeh et al. (2022) along with various energy industries as well as trade actors, and the effect it had on the efficiency of the power and power sector's operational capacity. To keep the electrical grid secure, improve its 24/7 availability, and achieve improved demand flexibility, advanced wireless networks are also crucial in green energy technology (Shahinzadeh et al., 2020; Shahinzadeh, Moradi, Gharehpetian, Nafisi, & Abedi, 2019b). Internet of energy (IoE) is another significant area of these advancements; it involves the use of modern electronic controls, sensors, switches, and intelligent metering system that can share data over IT networks (Ahmed, Rehan, Basit, Ahmad, et al., 2023; Ahmed et al., 2024; Khalid, Ahmed, AlMuhaini, & Savkin, 2024; Shahinzadeh, Moradi, Gharehpetian, Nafisi, & Abedi, 2019a). Several other recent developments are covered in Ahmed, Khardawi, AL-Ismael, and Khalid (2023), Bagherzadeh et al. (2020), Davoody-Beni et al. (2019), Mustafa, Ahmed, Basit, Alqahtani, and Khalid (2024), including cloud control networks, smart power management techniques, and grid controls based on AI-computation. These frameworks and technologies are promising. Over the past decade, data-driven digital technologies have grown in the energy and power industries, increasing their effect. Digitalization in the energy sector is improving. In recent years, energy firms have been more committed to digital technology adoption and use. Since 2014, global investment in digital infrastructure and software in the energy sector has grown by more than 20% annually (totaling USD 47 billion in 2016). The investment in digital technology was higher than that in the gas-fired power output (USD 34 billion) by 40% in 2016 (IEA, 2024a).

Therefore, a complete and unified understanding on the different ways digital technology might be used in energy and power systems is essential. Promoting wide social and policy dialogue requires this awareness, and understanding the merits and potential drawbacks of these applications is necessary to maximize their benefits. An extensive knowledge base helps establish powerful business models from technological discoveries (Weigel & Fischedick, 2019). The systematic categorization and uses of application domains are understudied. Previous studies have been focused on the current state and restrictions of specific technologies, thereby lacking a comprehensive and integrated examination of the future environment. Therefore, the precise categorization and thorough assessment of application areas are essential and can help numerous power sector players in academia, industry, business, and policy development make informed progress. Therefore, a brief, decisive, and well-structured evaluation of the digital technologies in energy and power systems is required. The presented Table 1 outlines the motivation for this study and the gaps in the previous reviews on digital transformation. Integrated information on AI, IoT, and blockchain computation for energy management applications will be provided to stakeholders, industrial practitioners, and academic researchers.

1.1. Global energy and power systems: An overview

Global energy business is transforming energy marketing, generation, and distribution. This sector is under pressure to reduce carbon emissions and manage power supply and demand mismatches in electrical networks. The major objective is switching from fossil fuels to carbon-neutral green energy (Engels, Kunkis, & Altstaedt, 2020). Carbon-neutral energy sources were used to obtain net-zero emissions. Significantly, 2021 witnessed a notable increase of nearly 7% in the generation of renewable power, thereby establishing an exceptional milestone of 522 TWh. The primary driver of this expansion was the widespread use of wind and solar photovoltaic (PV) technologies, accounting for almost 90% of the total growth. Notwithstanding this progress, the percentage of RESs in the overall power-generating portfolio increased slightly by 0.4 percentage points, thereby leading to a cumulative proportion of 28.7% in 2021 (IEA, 2024c). The sluggish expansion can be attributed to a substantial surge in worldwide electricity demand, reaching unprecedented levels as a result of the post-COVID-19 pandemic economic recovery. Droughts in many places hampered hydropower output. The rise in global energy consumption as the economy recovers from the COVID-19 pandemic heightened feelings of urgency. China contributed to over 50% of renewable electricity generation growth in 2021. In 2020, wind and solar PV capacities increased significantly, causing this phenomenon. This surge was boosted by the US and the EU, each contributing 8% to the overall increase. However, droughts in Brazil and Turkey reduced renewable electricity generation and hydropower output, and low wind speeds in the UK and Germany reduced wind-farm power generation outputs (Harbisher, 2023).

Producing the energy and power system involves basic challenges, including intermittency and power supply quality characteristics such as voltage dips, harmonics, surges, fluctuations, and frequency shifts. The stability and consistency issues in the existing grid topology are examined here. Volatility discourages utility companies from using renewable sources, constraining the efforts of the energy and electricity companies to reduce their carbon footprint over time. Enterprises must appropriately manage power-supply gaps to provide uninterrupted electricity to clients, and this can cause them to use more expensive and environmentally destructive diesel generators. The use of these alternative strategies can harm the operational efficiency and competitiveness of the firm. The unreliable power supply, especially in developing nations, is an inadequate power system infrastructure.

This necessitates a fundamental shift in understanding grid structure. Electrification and smart grid infrastructures can help optimize electricity networks, which can help integrate RESs more smoothly, particularly as these technologies become more sophisticated and efficient. Given the unpredictability of future technological breakthroughs, an AI-powered smart grid should reduce energy bills and power outages. This can be achieved by balancing the national and local energy demand and supply. Therefore, an inexpensive, reliable, eco-friendly, and carbon-free energy source is required. Digital technology can solve this problem.

1.2. Digital technologies

The International Energy Agency (IEA) characterizes digitalization as the escalating incorporation of information and communication technologies (ICTs), coupled with the convergence of virtual and physical realms. The German Association of Energy and Water Industries defines it as an interconnected network of applications, processes, and devices operating through the internet. Both these interpretations align with the concept that digitalization refers to transformation facilitated or hastened by digital technologies, thereby encompassing a blend of hardware and software, referred to as a cyber-physical system. Consequently, even the initial utilization of computers reliant on ICTs can be considered as a broader process of digitalization, in line with the understanding that it is not an entirely novel phenomenon.

Table 1
Research comparison with previous reviews.

Ref	Year	Core Concepts			Key contributions
		AI approach	IoT's approach	Blockchain approach	
Shi et al. (2020)	2020	✓	X	X	The potential uses of AI to the monitoring and sustainability of electrical grids are being investigated.
Qays, Ahmad, Abu-Siada, Hossain, and Yasmin (2023)	2023	X	✓	X	The IoT-assisted energy management systems are examined, including the primary innovations, applications, designs, and standards.
Al-Ali, Gupta, Zualkernan, and Das (2024)	2024	X	✓	X	Key results emphasize the relevance of adopting the proper big data software and hardware for each layer of big data design, providing a detailed description of the benefits and drawbacks associated with each of these resources.
SaberiKamarposhti et al. (2024)	2024	✓	X	X	Responsive and robust grids may be imagined with unsupervised control of energy, AI at the edge, and decentralization.
Koukaras et al. (2024)	2024	X	X	✓	This study highlights the importance of doing more research in order to overcome technological obstacles and improve the efficiency of blockchain computing in the context of green energy utilization in small and medium-sized enterprises (SGs).
Faheem, Al-Khasawneh, Khan, and Madni (2024)	2024	X	X	✓	This research offers large information which concentrate These statistics have been gathered from Solana blockchain-based computing.
This Work	2024	✓	✓	✓	This study provides a significant contribution to the larger academic discussion that is taking place about the use of digital transformation in order to develop energy ecosystems that are both intelligent and sustainable.

In recent decades, there has been a significant increase in data output across numerous industrial sectors due to the widespread use of the internet and advancements in ICT-compatible electronic gadgets. Based on projections provided by the IEA, it is anticipated that the quantity of internet traffic will reach a substantial 4.2 zillion bytes by the culmination of 2022, representing a fourfold increase compared to the data created in 2017. The expanding complexity of energy and power systems, particularly the integration of modern sensors and actuators with intelligent communication capabilities, highlights the increasing relevance of data inside these systems.

This comprehensive research spans three prominent digital technologies: AI, blockchain, and the IoT. Each of these technologies is centered around the utilization of data, with AI acting as the fundamental and all-encompassing component. Considerable research attention is devoted to these cutting-edge technologies. Several analyses have been conducted to examine these technologies, investigating their unique procedures and approaches that are utilized to tackle difficulties in the power sector. However, to the best of our current understanding, there has been limited attention paid to doing a thorough analysis of the combined utilization of these technologies.

The primary objective of this study is to conduct a thorough and application-specific evaluation of three emerging digital technologies within the realm of energy and power systems. These technologies are AI, the IoT, and blockchain. The focus is on classifying various energy and power domains for each technology and exploring the diverse applications that have been investigated over the past decade. The subsequent sections of this review are organized in the following manner: In the introductory part, we explore the methods utilized for the identification and selection of relevant and current literature. The following sections are devoted to the analysis of the practical uses of each individual technology within the framework of electric power systems. This paper entails a comprehensive examination of the practical implementations of AI, followed by a comparable evaluation of the IoTs and blockchain. In conclusion, we conduct an analysis of the growing patterns seen in each technology and ascertain possible areas that present opportunities for additional investigation in relation to their applications. The visual representation of the overall structure of this study may be observed in Fig. 1.

1.3. Scope and objectives

The increasing interest in the application of AI models in many industries has led to a surge in academic research in practical fields, particularly in the energy and power system sectors. Despite the existence of detailed assessments and overviews in specific areas such as artificial neural networks (ANNs), peer-to-peer (P2P) energy exchange, carbon credit trading facilitated by blockchain, and IoT applications in specific monitoring domains, the complex and extensive characteristics of these technologies limit their study to extremely specialized domains.

Thus there is a lack of comprehensive studies that provide a holistic view of the digital transformation in the energy and power sectors. A thorough and holistic understanding of this shift is crucial because of the inherent interconnection of various technologies, which does not enable them to function independently. Therefore, adopting a more comprehensive contextual framework is imperative for decision making and policy development. These technologies are not limited to operating at isolated nodes or components of the power system. Every application has a systemic effect permeating several regions of a grid. As a result, it is critical to have a comprehensive understanding in order to aid in the formulation of effective policies and the identification of prospective study areas.

Recognizing the gaps in existing surveys, this study aims to bridge the gap by delineating its scope as follows:

1. Providing a categorized examination of research trajectories, organized according to specific applications and technologies.
2. Discussing the role of the selected technologies in transforming the existing infrastructure.
3. Identifying the strong interdependence of technologies on each other and establishing their reliance on each other.
4. Establishing the 'system level' impact of the technology application.
5. Identifying research directions and trends in accordance with the aforementioned goals.

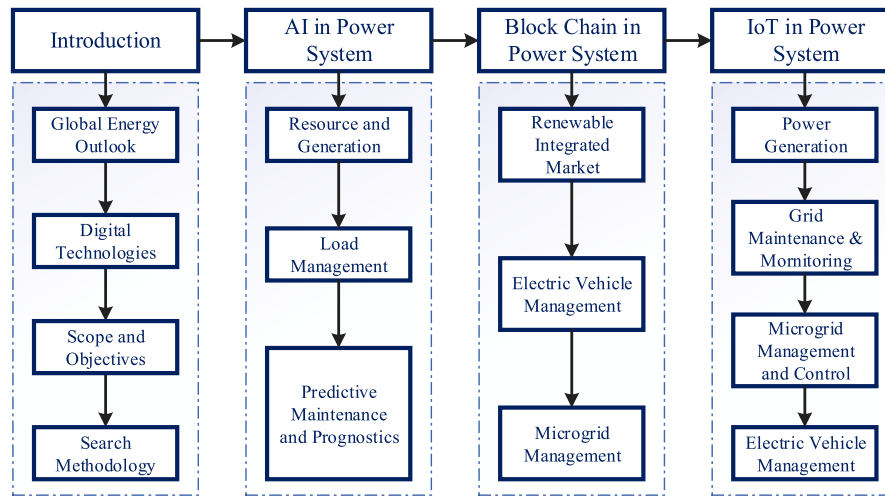


Fig. 1. Outline of the paper.

1.4. Search methodology

The procedure for choosing suitable material is dependent on and may vary based on the preferences of scholarly publications and search databases. This produces a fundamental bias when examining a certain scientific field. Taking this into consideration, we implemented a transparent and thorough strategy to guarantee the best possible results. Within the scholarly domain, the Scopus database search databases are widely regarded as a dependable and comprehensive reference. Scopus functioned as the primary channel via which we identified relevant articles. In order to refine our search, we employed specified phrases for every significant technology inside the scope of the research. We conducted a search using Scopus filters that specifically focused on predefined phrases found in titles, keywords, and abstracts. With the intention of offering a comprehensive examination of digital technology applications in power systems, the phrases "Artificial Intelligence", "Internet of Things", "Big Data Analytics", and "Blockchain" were consolidated with the term "Power Systems" in accordance with the objectives of the assessment. A thorough analysis was conducted on the abstracts, keywords, and content of the chosen publications in order to determine their direct relevance to the aims of the study. This paper aims to provide a comprehensive analysis of the search approach employed in the application of Artificial Intelligence. The aforementioned technique was later reproduced in order to investigate other technologies.

The primary aim of our study was to examine the practical implications of a specific technology referred to as "Artificial Intelligence applications in Power Systems". A complete collection of 1863 scientific articles was collected during the initial phase of data gathering. To ensure the timeliness of our review, we used a temporal filter that covered the time frame from 2018 to 2022. This resulted in a total of 1412 articles that were retained for further research. To minimize redundancy and maintain the relevance of established technological knowledge, we made the decision to exclude book chapters, reports, and similar publications from our research. Subsequently, the gathered results were refined to exclusively encompass "Articles", "Reviews", and "Conference Papers". In addition, our study was restricted to academic papers that had attained the "Final Published" phase, so excluding works that were still receiving peer review or were in the process of being published. In addition, we have refined the parameters of our investigation to include the domains of engineering, computer science, and energy. A total of 583 pertinent results were obtained within the defined parameters. After conducting a comprehensive evaluation of the abstracts, a total of 341 publications were identified as potentially relevant to the aims of our study.

For a comprehensive overview of other related technologies, please refer to Table 2 which provides a summary of their key aspects.

Table 2
Summary of search terms and retrievals.

Technology	Primary search term	No. of first-retrieved articles	Final selected
AI	"AI Implications"	1863	341
IoTs	"IoT Implications"	7,950	641
Block Chain	"Block Chain Implications"	985	340

2. Artificial intelligence in power systems

AI is a rapidly evolving and extensive technological topic within the world of technology.. AI serves as a comprehensive concept that incorporates several subdivisions, including machine learning (ML) and deep learning (DL), among other concepts. The aforementioned overall notion exhibits its influence in several practical domains, including computer vision, including both supervised and unsupervised learning techniques, as well as hybrid or semi-supervised learning strategies. These many classifications are encompassed in the broad field of AI, and are distinguished by their specific applications and operational capabilities.

Fundamentally, AI approaches aim to enhance the capabilities of computers or robots by simulating human cognitive processes. This enables machines to anticipate events, exercise control, and make decisions based on acquired knowledge from huge datasets used in AI model training. Robots can develop the ability to acquire information, tackle obstacles, forecast future situations, exert control over activities, and strategically plan in real-time environments through cognitive emulation (Ahmed, Hasan, et al., 2022; Ahmed, Rehan, Basit, & Hong, 2022; Akhter, Mekhilef, Mokhlis, & Mohamed Shah, 2019; Seyedmahmoudian et al., 2016).

The domain of AI offers significant opportunities for tackling the incorporation of RESs in urban planning, grid management, establishment of renewable incentives, standardization, and policy development. Several aspects of the incorporation of renewable energy can be improved by employing AI algorithms and conducting thorough data analysis. AI can provide accurate forecasts of energy demand, thereby aiding in the strategic planning and optimization of renewable energy-producing capacities. AI can evaluate geographical data, which aids in selecting ideal locations for implementing renewable energy systems. Consequently, this phenomenon has led to forming well-informed judgments in urban planning. In addition, AI algorithms can improve the efficiency of power networks by considering the intermittent characteristics of RESs; this optimization strategy results in increased stability

and efficiency within the grid system (Ahmed, Alvi, Ashfaq, et al., 2022; Ahmed, Irshad, et al., 2023; Ahmed, Rehan, Basit, Tufail, & Hong, 2023; Bedi, Goyal, Rajawat, Shaw, & Ghosh, 2022; Kanase-Patil et al., 2020).

Furthermore, AI assumes a crucial role in augmenting energy storage systems, efficiently utilizing clean energy sources and providing grid-related services. AI plays a crucial role in enabling the seamless integration and efficient management of green energy resources through its ability to provide continuous real-time monitoring and control of intelligent grid networks. In addition, they play a role in guaranteeing conformity with legal requirements, optimizing processes, and maintaining adherence to set norms and criteria pertaining to the integration of renewable energy. AI, with its ability to perform economic analysis, plays a crucial role in assessing the financial feasibility of renewable energy projects, contributing to the formulation of highly efficient investment plans. Furthermore, the use of AI-powered data analysis and the implementation of simulations that explore possible scenarios provide essential insights. These insights provide the basis for the development of evidence-based policy. The policies in question aim to streamline the integration of renewable energy by incorporating various aspects of urban planning, regulations, economics, and policy. The potential for redefining the integration of renewable energy through the combination of AI and quantum computing is substantial, promising more efficient, reliable, and environmentally sustainable power networks. The convergence of many aspects in this process contributes to the promotion of climate neutrality (Ahmad et al., 2021; Ahmed, Basit, et al., 2022; Ahmed, Rehan, Hong, & Basit, 2022; Ajagekar & You, 2022; Alvi et al., 2022; Șerban & Lytras, 2020). The substantial potential inherent in AI has spurred its rapid advancement, attracting considerable investment from both business sectors and academic communities. Consequently, the widespread adoption and influence of AI are on the rise across a diverse spectrum of disciplines. Notably, the realm of energy and power systems is experiencing a swift proliferation of AI applications. The prevailing landscape within energy and power systems is marked by a strong commitment to sustainability, accentuated by the drive to curtail carbon emissions, optimize energy consumption through intelligent means, foster energy conservation, and proficiently manage assets within the domain of energy and power systems. AI has adeptly evolved alongside the modern transformation, offering a diverse range of practical applications. For instance, one study (Ouahiba, Fatima, & Thafath, 2018) employs AI techniques to enhance occupant satisfaction within the broader context of smart buildings. In another context (Zahmatkesh & Al-Turjman, 2020), AI is harnessed to fortify the security of IoT systems. Moreover, AI technologies have been implemented in the context of green buildings for dynamic control of energy efficiency, as illustrated by a different study (Ouahiba et al., 2018).

Moreover, a variety of machine learning (ML) methods have been employed to forecast the availability of electricity resources, with a specific focus on intermittent renewable sources, as mentioned in Wang, Lei, Zhang, Zhou, and Peng (2019) and Wei, Li, Peng, Li, and Zeng (2019). This phenomenon goes beyond what it was originally meant to do. For example, the work (Ahmed, Rehan, Basit, Malik, et al., 2022; Jasiński, 2020) shows how machine learning models can be used to predict how power prices will change under dynamic pricing systems. Furthermore, there has been a notable increase in interest in applied AI across several sectors. The aforementioned factors encompass demand-side forecasting, demand-side load management, predictive maintenance practices, and power system stability preservation. The aforementioned studies (Ahmad, Zhang, & Yan, 2020; Ahmed, Alvi, Basit, et al., 2022; Ahmed, Rao, Shah, Alamzeb, & Khan, 2014; Ahmed, Rehan, & Iqbal, 2021; Ahmed, Rehan, Iqbal, & Ahn, 2023; Ali, Ahmed, & Sarwar, 2015; Badii et al., 2020; Basit, Tufail, Rehan, Riaz, & Ahmed, 2022; Dawn, Tiwari, & Goswami, 2019; Erixno & Abd Rahim, 2020; Mustafa et al., 2023; Ruhnau, Hennig, & Madlener, 2020; Shi et al., 2020; Wang, Lei, et al., 2019) demonstrate that these specific areas have become significant domains where AI techniques are extensively utilized.

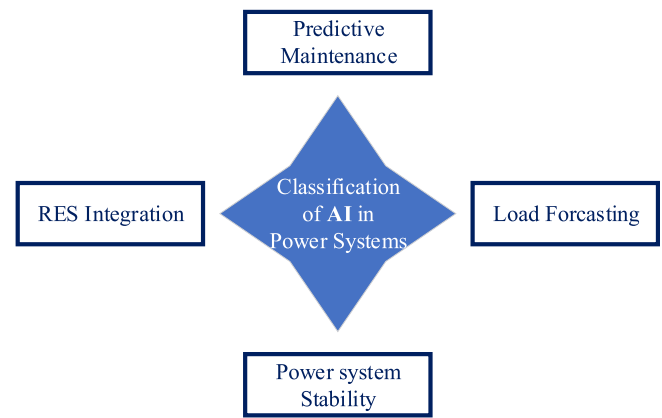


Fig. 2. Major application domains of AI in power systems.

The ensuing subsections offer a comprehensive analysis of AI applications, neatly categorized and expounded upon within each distinct category (as illustrated in Fig. 2). It is of utmost significance to underscore that the article selection process for this review was guided by its direct relevance to energy and power system applications. Therefore, we have intentionally excluded ancillary applications that involve the formulation or refinement of various AI models or algorithms from the scope of this review. Moreover, it is imperative to emphasize that our compilation comprises articles that may not overtly designate “AI” in their titles or abstracts. This suggests that our selection also includes articles that utilize ML, DL, and reinforcement learning techniques in the context of power systems.

2.1. Applications in power resource and generation

Three primary sectors commonly classify conventional power systems: generation, transmission, and distribution. The use of RES in the generating phase has resulted in notable transformations within these components. The aforementioned transition has given rise to several issues pertaining to the reliability of resources, the upkeep of infrastructure, and the effectiveness of operations. The rapid proliferation and extensive integration of clean energy sources into contemporary energy and power systems are particularly significant. The integration of renewable energy sources into the electricity grid presents complex problems, particularly in terms of grid stability caused by the intermittent nature of these sources. Although this integration aids in lowering carbon emissions, it also introduces complex problems that require attention. The variability in the output of various sources creates a level of unpredictability, adding complexity to grid management and supervision. The problem of instability becomes increasingly evident when energy users convert into prosumers amidst the rapid expansion of autonomous generation and the implementation of two-way power flow in distribution networks. In the present scenario, the use of AI techniques demonstrates significant benefits. Advanced AI techniques and algorithms have the potential to improve the feasibility of widespread adoption of RESs. By generating more accurate and reliable predictions of resource availability, we can enhance the overall reliability of the power grid. These technologies have the potential to maximize demand response, manage power distribution, and boost computational efficiency, thereby efficiently tackling the aforementioned difficulties.

Within the realm of integrating RES into power generation, AI methods are being utilized to enhance the arrangement of power facilities in solar, wind, and combined setups (depicted in Fig. 3). Improving hybrid renewable energy system (HRES) designs is important for making using renewable energy sources more effective and reliable (Caballero, Sauma, & Yanine, 2013). The use of intelligent AI algorithms has demonstrated their ability to deliver optimized outcomes with superior

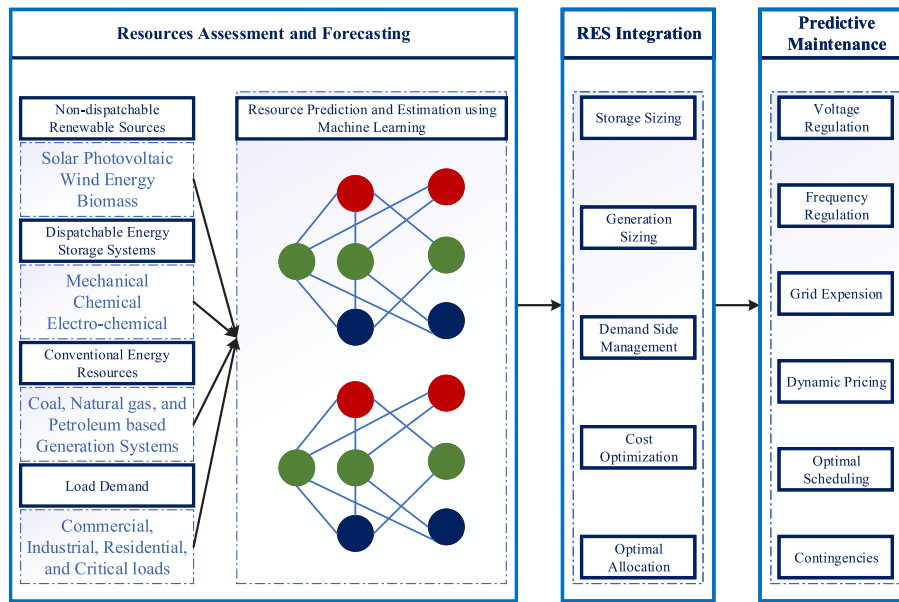


Fig. 3. Implementation and application of ML and AI for resource assessment.

computational efficiency, resulting in increased yields and enhanced operational effectiveness. For instance, in Mercado, Jiménez, and Quintero (2016), a genetic algorithm was applied for optimal sizing of HRES consisting of solar PV, wind turbines, and battery storage systems. Such hybrid systems are effective in remote areas and islands since they are more economical than the grid extension alternative. The developed genetic algorithm reduces cost and increases the system's reliability by calculating the optimal number and subsequently sizing the wind generators, PV modules, and accompanying battery storage.

Also, a careful and in-depth study of how AI is used to incorporate RES into a relevant framework (Al-Falahi, Jayasinghe, & Enshaie, 2017) revealed a clear upward trend in the creation of AI-based algorithms whose goal is to enhance design optimization. The analysis classifies optimization procedures into three main categories, including classical approaches, AI and ML techniques, and software utilities. The demand for contemporary approaches that focus on individual AI algorithms has increased in comparison to older procedures. This is mostly owing to the enhanced computing capabilities of these methodologies, which enable them to efficiently address complex problems. In the field of artificial intelligence algorithms, there is a preference for hybrid techniques rather than solitary ones since they provide advantages in terms of computing economy, execution speed, and precision.

The potential of AI applications to predict energy resource availability has been intensively researched in combination with design optimization. The prognostications in question possess wide-ranging utility because they can indirectly affect grid stability and assist in the further integration of RESs into the total generating mix through enhanced and precise resource projections. The intrinsic intermittency of renewable resources pose a significant challenge in precisely quantifying this uncertainty using standard approaches. Consequently, predictive AI methods become critical in driving the advancement of resource forecasting algorithms. These algorithms were designed with a focus on attaining an improved temporal resolution, enhanced accuracy, and decreased computational burden. For example, Hu, Cao, Wang, Li, and Lv (2018) employed an indirect predictive methodology to delineate cloud patterns by utilizing motion vectors. They utilized several approaches in digital image processing to quantitatively measure the dynamic characteristics of clouds. A radial basis function (RBF) neural network was trained using a model that employed data retrieved during image processing. The RBF model demonstrated a higher level of performance than the autoregressive model, thereby resulting in a significant enhancement in the accuracy of predictions.

Performance quantification depends on resource prediction, which helps gauge the effectiveness of RES installations, factoring in resource availability. In a previous study (Liu, Zhao, et al., 2018), a two-stage model was developed to quantify the performance indicators (PIs) of PV power outputs. The first stage involved integrating three types of neural networks—generalized regression neural network, extreme learning machine neural network, and Elman neural network—using the genetic algorithm-optimized back propagation method to create a weight-varying combination forecast mode model. Then, this model is used to generate deterministic point predictions (DPPs). In the second stage, a nonparametric kernel density estimation method was employed for estimating PIs based on the statistical distribution of errors in DPPs obtained in the first stage.

2.2. Applications in load and demand side forecasting

Predicting energy demand is extremely important for contemporary electricity markets, especially considering substantial regulatory transformations, such as unbundling. This suggests that the characteristics of electricity consumption are expected to undergo alterations for several reasons, including the integration of RES, proliferation of distributed energy sources, adoption of electric cars, implementation of environmental rules, and increasing prevalence of energy storage. Electricity forecasting has several applications in establishing a relevant framework for planning and maintaining adequate power quality standards, as indicated in Fig. 4.

In the context of short-term operations, it is imperative to cultivate precise demand projections, considering the generally stable nature of supply capacity. However, the task of accurately predicting future demand continues to present difficulties due to the influence of several factors, such as time of day, seasonality, and the distinction between working days and public holidays. Besides technical considerations, market and policy-making decision-makers need to evaluate future investments and their potential production capacity. These decisions are based on accurate predictions of future energy demand. Therefore, load or demand forecasting (LDF) has become a crucial component of standard grid processes as it enables real-time power production and load management. Typically, we categorize LDF into three types: short-term, moderate-term, and long-term forecasting, with forecast periods ranging from a few minutes to over a year. AI has the potential to solve this complex computational problem. Various AI techniques are widely

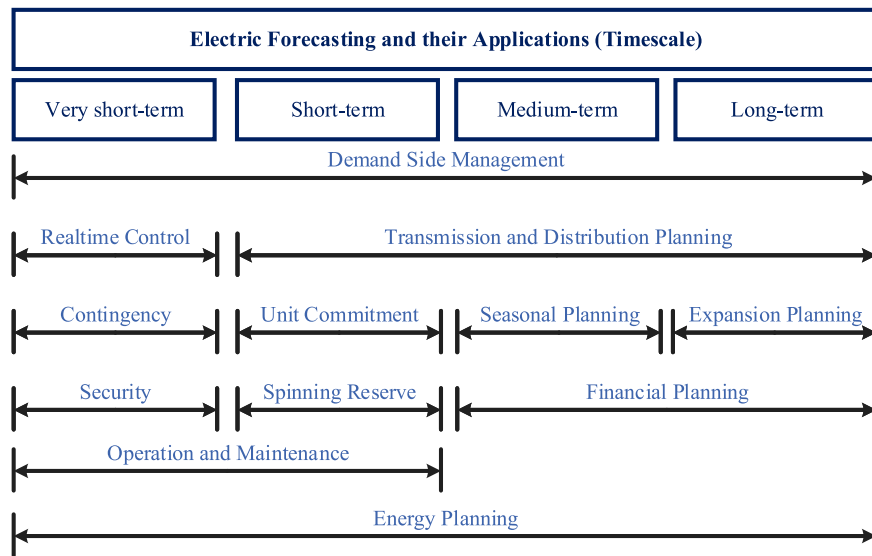


Fig. 4. The scope of forecasting in power system planning.

employed in load and demand forecasting. For example, a new hybrid data-driven technique called GA-BiGRU was proposed. GA-BiGRU is a combination of a genetic algorithm and a bidirectional gated recurrent unit. The prediction model's performance was compared with existing methods such as LSTM, GRU, and GA-GRU using metrics such as mean absolute performance error and root mean squared error. In a similar way, [Ahmed et al. \(2022\)](#) tried to improve the accuracy of electrical load demand forecasts by using LSTM and RNN architectures with deep neural networks. The paper explored the effect of seasonal resource variation on forecast errors.

The precision of AI-driven algorithms is heavily contingent upon the quality of their training. The dynamic nature of electric load profiles frequently introduces variations in the training dataset, leading to potential inaccuracies. Addressing these challenges, a novel strategy for load forecasting was presented in [Bendaoud, Farah, and Ahmed \(2022\)](#). This approach leverages Load Profiles (LPs) to mitigate the impact of load profile dynamism. To illustrate, an investigation into power consumption patterns in Algeria was conducted, revealing a spectrum of factors influencing demand. The introduction of seasonal fluctuations was accomplished by integrating hourly temperature profiles. The LP-centric forecasting methodology was executed across three distinct time scales – yearly, weekly, and daily – employing LP propagation techniques. To achieve this, predictive models were formulated for short- and medium-term load forecasting, harnessing various AI methodologies. Particularly noteworthy is the integration of a two-dimensional Convolutional Neural Network (CNN), which was employed for the first time in the domain of load forecasting. Likewise, within the context of this study, as presented in [Zhang, Tan, and Wei \(2020\)](#), a novel and adaptable hybrid model was put forth for the purpose of predicting short-term electricity prices. This innovative model was constructed through the fusion of variational mode decomposition (VMD), self-adjusting particle swarm optimization (SAPSO), seasonal autoregressive integrated moving average (SARIMA), and deep belief network (DBN) techniques. To gauge its performance, the efficacy of this newly introduced model was evaluated using datasets derived from electricity markets in the United States, Europe, and Australia.

In addition to the initial formulation of the model, researchers have directed their attention towards a comparative assessment of diverse models within practical applications. As an illustration, detailed in [Amber et al. \(2018\)](#), a meticulous comparison was carried out to evaluate the predictive capabilities of five distinct intelligent system techniques for forecasting electricity consumption in a government

building located in London, United Kingdom. The five methodologies under scrutiny encompassed Multiple Regression (MR), Genetic Programming (GP), Artificial Neural Networks (ANN), Deep Neural Networks (DNN), and Support Vector Machine (SVM). These prediction models were crafted utilizing five years' worth of empirical data, encompassing parameters such as solar radiation, temperature, wind speed, humidity, and weekday index. The outcomes revealed the superiority of ANN over the other four techniques, exhibiting a Mean Absolute Percentage Error (MAPE) of 6%, while MR, GP, SVM, and DNN displayed MAPE values of 8.5%, 8.7%, 9%, and 11%, respectively. A comprehensive representation of the machine learning framework employed in load prediction is illustrated in [Fig. 5](#).

Moreover, within the domain of empirical inquiry, researchers have explored hybrid AI models in order to improve computing efficiency, accuracy, and the general reliability of performance. In accordance with the objective of hybridization, the research described in [Kaytez \(2020\)](#) developed a hybrid model that combines the least-square support vector machine (SVM) with the autoregressive integrated moving average (ARIMA) approach. The present study introduced a hybridized approach aimed at forecasting future outcomes. This approach was then compared to alternative methods, such as a multiple linear regression model, a solitary autoregressive integrated moving average model, official projections, and previous research findings documented in the existing literature. The prognostications of the unique hybrid model encompassed the projection of Turkey's anticipated net power consumption up until the year 2022.

In a similar vein, researchers have devised an artificial intelligence-driven program with the purpose of forecasting wind power density ([Rodríguez, Florez-Tapia, Fontán, & Galarza, 2020](#)). The power density of wind may exhibit significant fluctuations, which can pose challenges in achieving a harmonious equilibrium between energy output and consumption, especially within conventional grid infrastructures. In order to tackle this matter, the suggested tool was designed with the objective of forecasting the forthcoming wind power density for a duration of 10 min, thereby enabling the enhancement of micro-grid control. The accuracy of the tool was assessed by conducting a comparison between the root mean square error of the predicted wind power density and the corresponding actual values. This evaluation was carried out throughout the validation period, which spanned from January 2017 to August 2017. The findings of the study indicate that the disparity between the observed and predicted wind power density was below 6% for about 81% of the days under consideration, with an average variation of 3.75%. Based on the results obtained

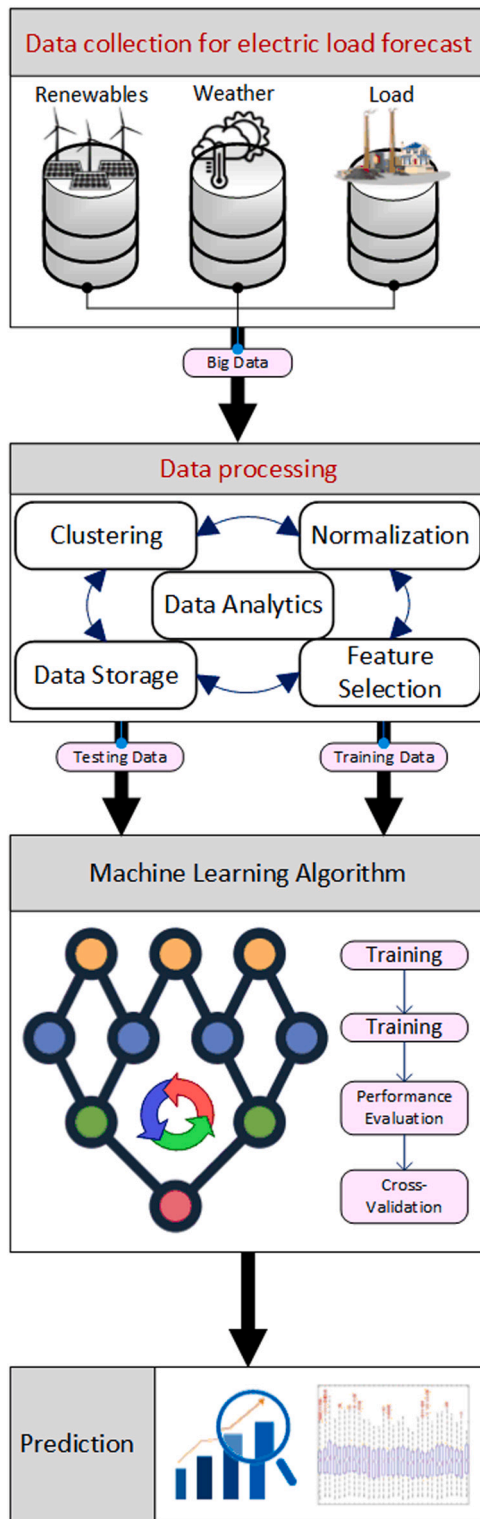


Fig. 5. The scope of forecasting in power system planning.

from the analysis, it was determined that the suggested forecasting model exhibited sufficient accuracy to warrant its integration into wind turbine systems, hence enhancing control strategy.

Similarly, in Bedi and Toshniwal (2019), a deep learning framework was designed to forecast electricity demand while considering long-term historical dependencies. The electricity consumption data for all

months was subjected to cluster analysis to create season-based segmented data. Then, load trend characterization was conducted to gain a deeper understanding of the metadata in each cluster. Subsequently, Long Short Term Memory (LSTM) multi-input multi-output models were trained to forecast electricity demand based on season, day, and interval data. The framework also incorporates moving window-based active learning to enhance prediction results. The proposed approach was tested on the electricity consumption data of Union Territory Chandigarh, India, and its performance was evaluated by comparing its results with those of ANN, RNN and Support Vector Regression models. Further, the study (Yuan, Farnham, Azuma, & Emura, 2018) presented ANN models to forecast seasonal, hourly electricity consumption for three areas of a university campus in Japan. Six parameters, including the day of the week, the hour of the day, hourly dry-bulb temperature, hourly relative humidity, hourly global irradiance, and previous hourly electricity consumption, were used as input variables for the ANN models. The models were developed using feed-forward ANN with Levenberg–Marquardt (LM) back-propagation algorithms. The correlation coefficient (R^2) and root mean square error (RMSE) was used to evaluate the accuracy of the proposed ANN models. Results showed that the R^2 between actual measurements and ANN model predictions ranged from 0.96 to 0.99 in the training stage and from 0.95 to 0.99 in the testing stage. The RMSE was found to be the largest in the Science and Technology area of the university campus, followed by the Humanities College area. The Old Liberal Arts area had the smallest RMSE and difference.

2.3. Predictive maintenance and prognostic applications

Industry 4.0, the latest phase of industrial evolution, closely aligns with the notion of PM. This concept entails the continuous monitoring of the operational performance and condition of the equipment during its routine functioning, to diminish the potential for breakdowns. Predominantly observed in the manufacturing sector, where electro-mechanical systems play a prominent role, PM has emerged as a strategy that capitalizes on acquiring and interpreting data from diverse sources such as sensors, machinery, and switches. This methodology aims to preemptively prevent equipment malfunctions by implementing sophisticated data-driven models.

The integration of AI and ML techniques elevates the level of automation in determining optimal maintenance and replacement schedules. This is achieved by tapping into insights gathered from past maintenance instances, various failure patterns, system durability, and associated costs of failure. Moreover, the current trajectory of research and development is concentrated on harnessing AI and ML models to bolster the security of energy networks. This involves setting up preemptive mechanisms to detect unexpected cyberattacks or physical disturbances within the energy infrastructure. In this segment, we provide an extensive overview of the diverse ways AI is employed in enhancing the maintenance processes of energy power systems.

Electro-mechanical machinery makes up a significant portion of the power generation infrastructure. Although the current energy transition may result in a decrease in the frequency of these electro-mechanical units, we do not expect their significance to decrease significantly. Consequently, the maintenance of these electro-mechanical systems retains its paramount significance within the energy and power sectors. Given its substantial reliance on such systems, the realm of hydropower generation has emerged as a pivotal domain for the application of prognostic AI. For instance, in the work denoted by Zhu et al. (2014), an innovative Kernel Independent Component Analysis (KICA-PCA) model was introduced for fault diagnosis. This model harmonizes the advantages inherent to Principal Component Analysis (PCA) and Independent Component Analysis (ICA). The procedure commences by transforming nonlinear sensor data into a higher-dimensional linear representation using a kernel function. Subsequently, ICA is employed to distill relevant features from vibration data. The identification of

vibration-related faults is facilitated through the application of statistical metrics such as Hotelling's T² Statistics and the Squared Prediction Error (SPE).

Furthermore, recent research has examined the application of cloud computing to address the computational requirements associated with predictive maintenance in hydropower plants. The article by the authors (Kande, Isaksson, Thottappillil, & Taylor, 2017) provides a thorough examination of the many strategies employed in the implementation of monitoring systems into automated industrial systems. These strategies encompass both on-site and distant integration, utilizing cloud-based IoT devices. Although these strategies demonstrated effectiveness, they need more scrutiny with regards to their security implications.

In order to ensure the efficacy of future Condition Based Maintenance (CBM) systems for rotating electrical devices, it is imperative that they possess the qualities of cost-effectiveness, security, and the ability to be deployed on a broad scale. Nevertheless, the use of cloud-based installations for these systems may lead to potential issues such as latency and network congestion. In order to tackle these concerns, a novel Predictive Maintenance (PHM) platform was suggested by Xiao, Zhang, Tian, Huang, and Fu (2018), which used fog computing and docker container technologies. The fog computing architecture has successfully addressed the challenges associated with time delays and network congestion, resulting in improved real-time processing capabilities inside a cloud-based architecture. The fog-computing-based fault diagnosis system was effectively deployed at the Zhen Toubu Hydropower Plant.

Moreover, there have been suggested deep learning-based systems for the purpose of predictive maintenance in hydropower plants. In the study conducted by the researchers cited as Zhang et al. (2019), DNNs were utilized to detect anomalies and faults in multivariate time series data. The researchers employed the building of multi-scale signature matrices to characterize the different degrees of numerous system states at various time steps. The matrices were employed for the purpose of encoding inter-sensor correlations through the utilization of a convolutional encoder. Additionally, temporal patterns were collected by means of an attention-based convolutional LSTM network. The temporal patterns and feature maps were subsequently utilized as input for a convolutional decoder in order to reconstitute the input signature matrices. Anomalies were identified by analyzing the residual signature matrices. The results of experiments conducted on a synthetic power plant data set demonstrated that the suggested framework exhibited superior performance compared to the current state-of-the-art baseline prognostic approaches.

In addition, there has been a deliberate exploration of systems dedicated to monitoring the condition of individual components. These endeavors have been particularly directed towards detecting faults in bearing and gearbox components within hydro power installations, achieved by scrutinizing vibration data. In a specific example, highlighted by Wang, Tsui, and Miao (2017), an exhaustive examination was undertaken encompassing the monitoring of bearing and gear health via vibration signals. The study's authors delved into the correlation between indicators of health and the Remaining Useful Life (RUL) of these components. Through an investigation of data derived from experiments simulating both operational degradation and accelerated bearing deterioration, the authors identified and evaluated various vibration-based health indicators, as well as the associated challenges encountered in bearing and gearbox systems. In a similar vein, the authors in Pino, Ribas, and Guimarães (2018) devised a methodology to assess the deterioration of guide bearings in hydroelectric projects. This approach leverages the wavelet packet transform (WPT) and Hidden Markov Model (HMM) in conjunction with orbit curves. The system's purpose was to mitigate the increasing deterioration of mechanical elements in medium-scale hydroelectric projects. This deterioration occurs when the turbine operates in a transient condition at maximum loads, resulting in amplified mechanical vibrations in the runner's bearings.

The vibration measurements obtained during the transient state were subjected to filtering and analysis using WPT. This analysis enabled the establishment of a correlation between the maximum displaced orbit curves and the number of operational hours. The HMM was subsequently employed to model the degradation process and forecast the series of maximum orbit curve displacements. The method successfully mitigated the likelihood of failure and minimized maintenance costs through the utilization of probability calculations to assess the degradation levels of the guide bearings. Comparable models were identified in the prognostication of wind turbines as well. Due to their strong resemblance to the electro-mechanical systems found in hydro power infrastructure, these systems were classified under the same conceptual framework.

Furthermore, utility-scale solar power parks have used AI for maintenance and defect detection in addition to electro-mechanical maintenance.

In conventional practice, PV systems were monitored to assess module and array performance, grid stability, power factors, and islanding mitigation. Recent research has prioritized the development of PV power output monitoring systems with the aim of effectively detecting and addressing arc-fault and ground-fault occurrences (Karatepe et al., 2011). The primary goal of PV monitoring systems is to identify and analyze deviations or irregularities from standard operational conditions. To achieve this goal, researchers have developed several models that utilize temperature and irradiance data from sensors or meteorological and satellite data sources to estimate the anticipated power output (Drews et al., 2007; Karatepe et al., 2011). The anticipated power is afterwards juxtaposed with the factual output in order to scrutinize any alterations in system behavior.

In addition, a previous study Riley and Johnson (2012) introduced an ANN architecture as a means to monitor the health of PV systems, quantify degradation, and establish maintenance timetables. During experimental trials conducted on research facilities, the system demonstrated effectiveness in identifying system malfunctions that resulted in a decrease in power production. A separate investigation conducted by the researchers referenced as Livera et al. (2022) introduced a machine learning-driven framework for the purpose of PV performance modeling and failure diagnostics. The architectural design incorporated a predictive model that was trained using a limited amount of on-site data, as well as functional processes for the verification of data quality. The classifier underwent training using an enhanced method of supervised learning. The outcomes indicated that the model achieved a notable degree of accuracy, with a sensitivity of 83.91% for synthetic power-loss events and 97.99% for field-emulated failures in the test-bench PV system. Thermal imaging has emerged as a dependable tool for identifying Potential Induced Degradation (PID) in modules, even in new PV installations where significant power loss may occur. An illustrative study introduced a forward-looking system rooted in thermographic imagery to identify power loss in PID-affected silicon solar modules. This innovative approach employed remote-controlled multi-copters to carry out inspections without the need for disassembly, as detailed in Kaden, Lammers, and Möller (2015).

Furthermore, the evaluation of energy storage system performance and degradation has been undertaken by researchers as outlined in Karoui et al. (2018). They devised a sophisticated tool named "A4" that capitalizes on advanced data analysis techniques to enhance system efficiency and availability.

Additionally, the realm of AI-based maintenance applications has extended to micro-grids. Ensuring the economic viability of micro-grids, equipped with diverse and widely dispersed energy sources, necessitates heightened reliability and minimized maintenance costs. This has spurred the demand for real-time condition monitoring and health prognosis systems. The Performance Ratio (PR), a pivotal metric gauging the actual energy generation against the rated capacity of solar cells, plays a critical role in determining a solar PV system's reliability.

Table 3

Classified summary of applications of AI in energy and power systems.

Application classification	Core concept	Challenge area	Year	Ref.
Integration	Development of a method of optimal business design for small hybrid-energy system (grid-connected) constituting of PV and wind turbines, ensuring reliable power supply.	Minimizing life cycle cost and improving loss of power supply this is measured in terms of loss of power supply probability	2013	Caballero et al. (2013)
of	Optimal sizing of HRES consisting of Solar PV, Wind Turbines and Battery Storage through the application of genetic algorithm.	Remote and off-grid electrification	2016	Mercado et al. (2016)
Renewable	Critical analysis of size optimization methodologies, as well as a critical comparison of single algorithms, hybrid algorithms, and software tools used for sizing	Standalone solar and wind HRES	2017	Al-Falahi et al. (2017)
Energy	Development of model for prediction of photovoltaic power generation in ultra-short term range. The three stage model employs motion vector, digital image processing and radial base function (RBF)	RES resource prediction	2018	Hu et al. (2018)
Sources	Development of a model to quantify the performance indicators (PIs) of photovoltaic (PV) resource using artificial neural networks (ANNs)	RES resource prediction	2018	Liu, Zhao, et al. (2018)
	AI based tool to predict wind power density, in order to develop a control and operation strategy of microgrids with wind installations.	Microgrid control and RES resource prediction	2020	Rodríguez et al. (2020)
	Develop an evolutionary DL based hybrid approach for short term load forecasting (STLF). Owing to limitations of singular methods, integrated technique GA-BiGRU was employed.	Demand side management and load forecasting	2022	Inteha et al. (2022)
Load	Development of a model based on Deep Neural Networks for short term load forecasting, applying activation functions such as sigmoid and ReLU. Reliability testing of models using RMSE, MAPE and RSE.	Load forecasting and daily operational planning	2022	Ahmed et al. (2022)
and	Load Profile (LP) based ML forecasting method. The forecasting is done at three different levels of LP-propagation namely annual, weekly and daily. STLF and MTLF techniques were developed using AI methods, specifically 2-D CNN.	Load profile analysis and prediction	2022	Bendaoud et al. (2022)
Demand	Hybrid adaptive model combining Variational Mode Decomposition (VMD), ARIMA, and PSO and Deep Belief Network to reliably predict electricity prices in short term.	Electricity price forecasting for dynamic systems	2020	Zhang et al. (2020)
Forecasting	Critical assessment and comparison of predictive ability of five intelligent techniques including Multiple regression, DNN, ANN, Genetic Programming and SVM.	Reliability of forecasting	2018	Amber et al. (2018)
	Hybrid model of electricity consumption based upon least-square SVM and ARIMA. A comparison with single ARIMA based model was performed	Net electricity consumption forecasting and power system planning	2020	Kaytez (2020)
	DL based load forecasting incorporating historical dependencies and cluster analysis. Long Short Term Memory network multi-input multi-output models are trained to forecast electricity demand based upon the season, day and interval data.	Load forecasting	2019	Bedi and Toshniwal (2019)
	A critical review of the condition monitoring techniques and the recent advancements in intelligent CBM.	Machine monitoring	2017	Kande et al. (2017)
Prognostics	Fog computing based hydropower equipment prognostics and health management structure. Improved real-time processing capacity and cloud based prognostics architecture was developed.	Hydropower equipment health management	2018	Xiao et al. (2018)

To this end, [Bandong, Leksono, Purwarianti, and Joelianto \(2019\)](#) harnessed machine learning methodologies to predict and estimate PR for Solar Power Plants, facilitating informed decisions about maintenance and the replacement of underperforming cells. Their approach entailed a model based on PCA and SVM, utilizing three years of weather data to yield superior outcomes compared to conventional methods like Multiple Linear Regression (MLR).

Deep learning techniques, encompassing ANNs, RNNs, and LSTM networks, have been effectively employed for the prognosis of solar energy systems, as exemplified in [Correa-Jullian, Cardemil, Droguett, and Behzad \(2020\)](#). Notably, LSTM demonstrated enhanced accuracy over DNNs and outperformed RNNs in predicting temperature sequences. These deep learning models collectively outperformed conventional approaches. Similarly, in [Kelker et al. \(2019\)](#), ANNs were trained using real-world measurements to forecast the output power of a PV system.

A comprehensive tabulation of AI applications in Energy and Power Systems can be found in [Table 3](#), and a representative depiction of the virtual component model formulation architecture is depicted in [Fig. 6](#).

3. Block chain in power systems

Blockchain is a decentralized data storage system that eliminates the necessity of a middleman in contractual agreements or transactions, operating on a peer-to-peer (P2P) basis. The principles of decentralized data storage and sharing form the foundation of the system. People commonly refer to a decentralized database as a “distributed ledger”. Cryptographic techniques connect the blocks of recorded data that make up the ledger. This design imparts the ledger with inherent resilience against unauthorized access or cyber attacks, rendering it a highly secure method for exchanging information, value, or contractual agreements. The decentralized database is dispersed across a network of computers, referred to as nodes. Every participating node in the network holds a duplicate of every transaction or value exchange and assumes the role of a verifier for these transactions. The use of a multi-agent verification system mitigates the potential for single-point failure that is commonly associated with conventional third-party

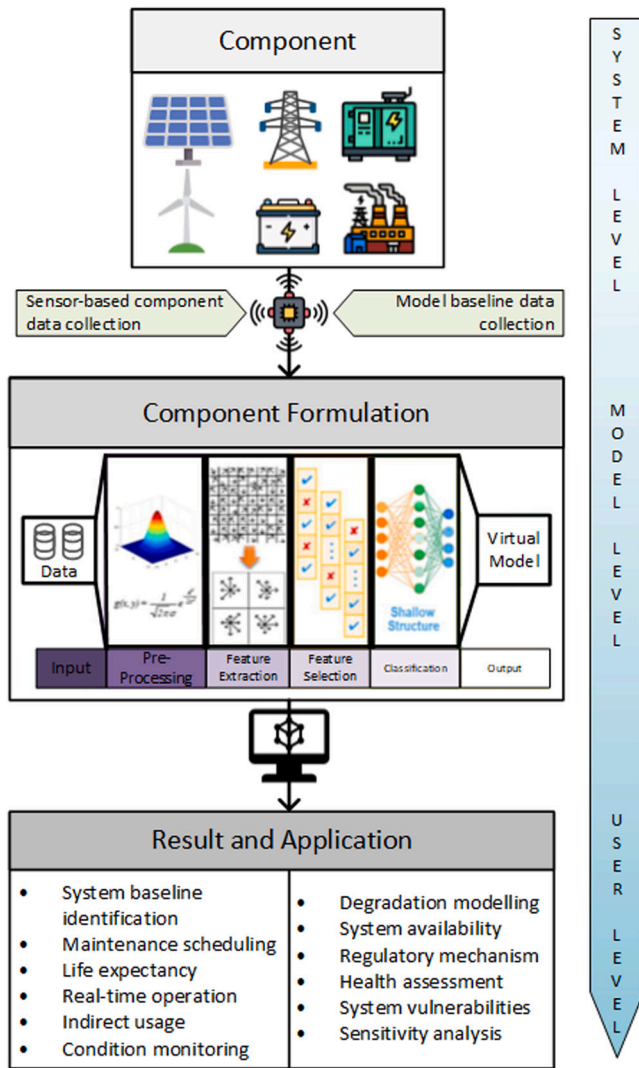


Fig. 6. Typical flowchart for development of virtual component models.

insured or centralized database transactions (Wei et al., 2019). One of the most notable attributes of blockchain technology is its inherent immutability. Once stored in a data structure, editing, altering, or substituting data becomes a complex task. In order to make any updates to a specific information block, it is imperative that all checking nodes within the network achieve an agreement. The elimination of a third party reduces transaction costs and improves the process's efficiency and transparency. In addition, the decentralized nature of blockchain technology facilitates transparency since it enables all network participants to view and authenticate the data. Fig. 8 provides a concise depiction of the blockchain. Each block, as shown in the schematic design above, consists of a header and a body. Both of these components are responsible for storing various sorts of information. In a blockchain system, the header of a block contains hash codes for both the preceding block and the current block. This is intended to establish a linkage with the preceding segment. Moreover, the block header contains the timestamp of the block. A timestamp refers to the specific time at which a certain block is generated. The block's content encompasses the transaction record, together with the public and private keys of the recipient, as well as the public key of the sender, which corresponds to the preceding block. The keys might be seen as distinct authentication codes that enable users to get entry to the information. Public keys serve as the distinct identification mechanism

for a node that is accessible to all other nodes. In traditional transactional and contractual frameworks, the oversight and assurance of agreements are typically entrusted to external entities, such as banks in financial systems and electric utilities in power systems. Nevertheless, it is important to note that the blockchain verification process requires active participation from all nodes or participants. This is because the network disseminates the crucial data, making it accessible to any individual node or participant. Consensus algorithms are crucial in this context. In order to include a new block, and therefore a transaction, into the blockchain, it is necessary to achieve consensus among a predetermined number of nodes. The consensus algorithm plays a crucial role in guaranteeing the integrity and accuracy of transactions or value exchanges. This is achieved through multi-agent verification, which greatly reduces the likelihood of fraudulent or malfunctioning nodes. Numerous consensus algorithms are employed in several blockchain protocols, each exhibiting distinct merits and drawbacks (Fig. 8). Several widely used consensus mechanisms in blockchain technology include Proof of Work (PoW), Proof of Stake (PoS), Proof of Authority (PoAU), and Practical Byzantine Fault Tolerance (PBFT). The applicability of these consensus techniques varies across different sorts of applications. Proof of Work (PoW) and Proof of Stake (PoS) are often employed in public blockchains, but Proof of Authority (PoAU) is better suited for private or hybrid blockchains. Proof of Authority (PoAU) diminishes the attribute of trustlessness and amplifies the vulnerability of single-point failure in a broader sense. Moreover, it has a higher susceptibility to malevolent intrusions. However, it is deemed appropriate for enclosed systems. The system exhibits a shorter decision time compared to alternative approaches and restricts participation in consensus verification exclusively to approved nodes. The micro-grid peer-to-peer exchange in the electric power industry is a very influential and impactful application (see Fig. 7).

It is important to note that blockchain functions as a trust protocol because of its basic design, thereby eliminating the need for trust. There are several widely recognized protocols that have gained worldwide popularity, such as Bitcoin, Ethereum, and Hyperledger.

3.1. Applications of block chain in power systems

Block chain technology has been picking up the interest of power system researchers and practitioners alike. The energy sector is shifting towards decentralized systems and deregulated markets, and blockchain technology has the potential to address various issues in this field due to its transparency, decentralization, anonymity, and reliability. The introduction of new equipment and energy sources, along with bidirectional communication among stakeholders, has resulted in the creation of a SG. The integration of RES, energy storage systems, EVs, DG, and MG in the SG renders the traditional centralized power management model unattractive due to cost, security, and control drawbacks. In this section, we elaborate on the applications of blockchain in energy and power systems. In Zakaret et al. (2022), a combinational method was proposed for building root of trust in IoT platforms. Secure Element (SE) and block chain technologies were proposed to be combined to make a more secure distributed data storage in a smart grid setting. The approach was validated using an energy smart metering gateway which proved to produce valid results. The SE is introduced as provision for a secure root of trust. The overall aim of the study is to enhance the security of the P2P energy trading, by filling the gap in security at the point of generation. Similarly, in Zhang et al. (2022) different issues in hierarchical data management structure of electric power systems were addressed through a block chain based digital asset management framework. The paper optimized security authentication protocols within the asset management protocol. The study also optimized consensus process, data sharing and storage mechanism to ensure holistic integrity of the block chain application.

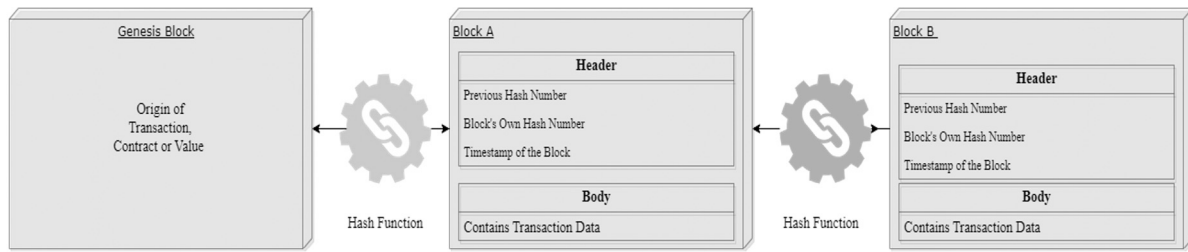


Fig. 7. Basic architecture of block-chain.

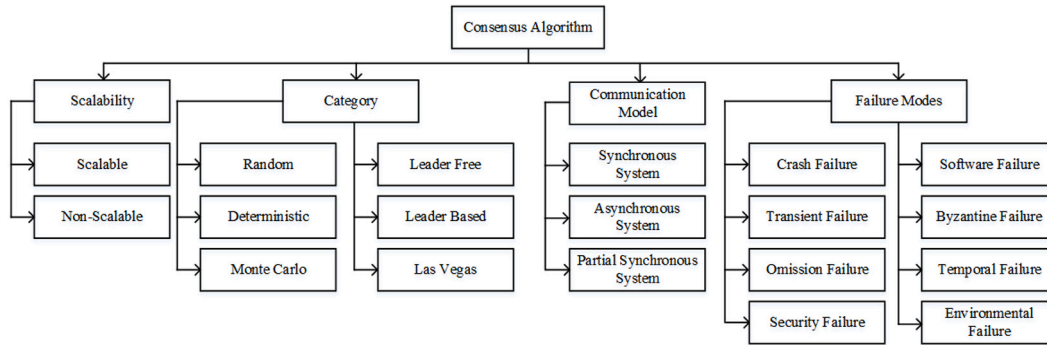


Fig. 8. Basic architecture of block-chain (Chaudhry & Yousaf, 2018).

3.2. Electric vehicle management

Bringing the smart contract features of block chain to use, in [Su et al. \(2018\)](#) a contract-based energy block chain was developed for secure EV charging mechanism. A permissioned energy is block-chain system is used for implementing for charging services. For the consensus building, a reputation-based delegated Byzantine fault tolerance consensus algorithm is proposed for enhanced efficiency of the process. Further, the study also designed and analyzed. An energy allocation mechanism is also developed for allocating renewable resources to EV charging. Similarly, other research reviewed, also focused on the developing secure Vehicular Energy Networks based on block chain technology. In [Wang, Su, and Zhang \(2019\)](#), the authors developed a block-chain based permissioned energy block-chain system. The consensus protocol employed was proof of reputation(PoR), with reputation derived from trust computation index. Moreover, an incentive model was developed in line with the pricing mechanism, aimed at co-operative promotion of EVs and RE. Further, [Liu, Chai, Zhang, Lau, and Chen \(2018\)](#) is concerned with reducing the power fluctuation in grid, caused by the large scale penetration of EV charging infrastructure. An adaptive block-chain-based EV participation framework. It uses Iceberg order execution algorithm to formulate an efficient EV charging and discharging schedule. In comparison to genetic algorithms, the framework performs more efficiently, lowering the power fluctuation and costs of EV charging.

3.3. Microgrid management

In the EV industry, blockchain technology is gaining increasing traction as a solution to various challenges arising within the expanding realm of MGs. As these MGs proliferate, they present unique complexities that necessitate innovative solutions.

For example, in [Wang, Su, and Zhang \(2019\)](#), a novel co-simulation was discussed, integrating electricity distribution networks and a blockchain-based energy market or transactive energy platform. This endeavor involved modeling the power network and constructing a digital twin model, serving as the physical layer of the blockchain-based energy trading system. The resulting decentralized application

was tested in real-time within a housing community in Qatar, utilizing the Hyperledger Fabric blockchain protocol, and exhibited promising outcomes.

Furthermore, the application of blockchain to address issues related to distributed generation was noted. In [Asefi, Madhwal, Yanovich, and Gryazina \(2021\)](#), the integration of distributed state estimators with a blockchain-powered data exchange platform was proposed. The model accounted for the synchronization of power and data transfers, mimicking real power system dynamics. Numerical analysis verified the effectiveness of the proposed model in meeting real-time requirements while ensuring robust data security and reliability for state estimation processes. Similarly, a system for intelligent terminal-based power grid operation, combining IoT and blockchain technology, was developed as outlined in [Dong, Ji, Shen, Hu, and Di \(2022\)](#).

The surge in distributed energy generation and consumption prompted a study on the concept of a virtual power plant (VPP) introduced in [Ma, Tu, Zhou, and Xu \(2022\)](#). This VPP aggregates power resources for semi-centralized transactions, and a blockchain-based model was established to align with the characteristics of blockchain, such as security, decentralization, and transparency. To address challenges in managing distributed energy systems, a game theory-based model was presented in [Noor, Yang, Guo, van Dam, and Wang \(2018\)](#) for demand-side management, incorporating blockchain technology to ensure secure implementation.

In [Li, Bahramirad, Paaso, Yan, and Shahidehpour \(2019\)](#), a scheme for networked micro-grids was discussed, leveraging blockchain technology to optimize both financial and physical operations. This involved the development of a set of interoperable blockchain protocols with self-enforcing smart contracts, facilitating energy and financial flows among participating micro-grids. A novel microgrid restoration framework based on blockchain technology was proposed in [Mishra and Jena \(2022\)](#), encompassing a two-stage restoration process. Stable blockchain links were established with grid-forming Distributed Energy Resources (DERs), followed by load assignment based on priority levels. Researchers in [Yu et al. \(2022\)](#) introduced a dual-blockchain system for peer-to-peer (P2P) energy trading, comprising a primary blockchain for transaction data storage and a secondary blockchain for compute-intensive tasks. An Improved Optimistic Rollup (IOR) mechanism was

Table 4

Classified summary of block-chain advances in power system.

Application classification	Core concept	Block-chain Protocol	Year	Ref.
Electric Vehicles	A contract-based energy block chain was developed for secure EV charging mechanism. A permissioned energy is block-chain system is used for implementing for charging services. For the consensus building, a reputation-based delegated Byzantine fault tolerance consensus algorithm is proposed	Permissioned BFT	2022	Su et al. (2018)
	Developed a block-chain based permissioned energy block-chain system. The consensus protocol employed was proof of reputation (PoR), with reputation derived from trust computation index. Moreover, an incentive model was developed.	Permissioned BFT	2022	Wang, Su, and Zhang (2019)
	An adaptive block-chain-based EV participation framework. It uses Iceberg order execution algorithm to formulate an efficient EV charging and discharging schedule to mitigate power fluctuation in the grid	Ethereum	2021	Liu, Chai, et al. (2018)
	Implemented IoT-BC based EV charging scheme at public chargers	Ethereum	2019	Martins, Ferreira, Monteiro, Afonso, and Afonso (2019)
	Investor focused BC based EV charging scheme to maximize profits	DBFT	2019	Sun et al. (2018)
Peer-to-Peer	Designed a BC based battery swapping mechanism.	Ethereum	2018	Kim et al. (2019)
	Developed a IoT-BC based prototype for battery swapping for EV	Ethereum	2018	Yahaya et al. (2020)
	Develop BC for EVs with lesser computational requirements	Hyper Ledger	2019	Park, Chitchyan, Angelopoulou, and Murkin (2019)
	Improved Optimistic Rollup (IOR) mechanism is proposed to facilitate the co-operation of the dual blockchains. Through a specific 'executor-validator' mechanism and a group of specific smart contracts with different functions, the IOR mechanism can effectively accommodate the dual blockchains to execute the P2P energy trading transactions in a computational efficient manner	Ethereum	2022	Yu, Luo, Pu, Zhao, and Ranzi (2022)
	Develop a blockchain-based LEM in a small community that allows prosumers and consumers to trade electricity	Ethereum	2020	Wang, Xu, Xu, and Shao (2018)
Energy Contracting	Develop a blockchain-based mechanism for multi-directional P2P energy trading between microgrids. The results showed that the proposed mechanism increased RESS utilization and microgrids' profits.	Ethereum	2018	Hartmann and Thomas (2020)
	Investigate IOTA DLT platform for P2P energy trading and overcoming blockchain weaknesses. The simulation results showed the feasibility of IOTA for creating platforms for P2P energy trading	Ethereum	2019	Khaqqi, Sikorski, Hadinoto, and Kraft (2018)
Emission Trading and Renewable	Propose a blockchain-based ETS. In this ETS, no central organization is needed to manage bids and offers of participants. The proposed ETS can enhance traditional ETSS efficiency and tackle management and fraud issues.	MultiChain	2018	Castellanos, Coll-Mayor, and Notholt (2017)
	Investigate the application of blockchain technology in existing carbon markets, focusing on the Australian carbon market as a case study. The study presented the design process. The study showed that blockchain could improve the carbon market performance and efficiency, reduce transaction costs, and increase transparency and equity.	Not Specified	2020	Hu et al. (2018)
Energy Sources	Propose a blockchain-based RECS market that allows direct transactions between producers and consumers. The results showed that blockchain could be used to create a robust platform for RECS trading with low operating costs.	Ethereum	2017	Hua et al. (2018)

proposed to optimize the co-operation of the dual blockchains for efficient P2P energy trading transactions.

Smart distribution grids, incorporating fault self-healing capabilities, were explored in [Gebraselase, Adrah, Amare, Helvik, and Heegaard \(2022\)](#). A novel concept employing blockchain as a second-tier security mechanism was introduced to support time-critical self-healing operations, demonstrating its effectiveness through simulation studies.

A comprehensive compilation of blockchain applications is presented in [Table 4](#) as discussed in [Castellanos et al. \(2017\)](#), [Hartmann and Thomas \(2020\)](#), [Hua et al. \(2018\)](#), [Khaqqi et al. \(2018\)](#), [Kim et al. \(2019\)](#), [Martins et al. \(2019\)](#), [Park et al. \(2019\)](#), [Su et al. \(2018\)](#), [Sun et al. \(2018\)](#), [Wang et al. \(2018\)](#), [Yahaya et al. \(2020\)](#).

4. IoT-based technologies in power systems

The IoTs is conceptualized as a technological framework designed to establish connections and networks between different objects, often

referred to as 'things.' This framework operates through an Internet communication protocol, facilitating the exchange of data and communication among various intelligent devices. Its purpose is to intelligently fulfill monitoring and operational management objectives. The IoT, sometimes referred to as the Industrial Internet, envisions a global network where interconnected equipment and objects form a cohesive system ([Lee & Lee, 2015](#); [Wang, Li, Liu, & Wang, 2014](#)). At its core, the IoT encompasses three fundamental elements that synergize to create highly automated systems capable of both internal communication and interaction with external entities, primarily humans. These elements are typically labeled as 'things,' 'internet,' and 'semantic' or 'application' features. The 'things'-oriented aspect revolves around intelligent devices positioned at the interface between the system, external elements, and subsystems. These devices enable data collection by the system or its subsystems. Included in this category are intelligent sensors, actuators, cameras supported by computer vision

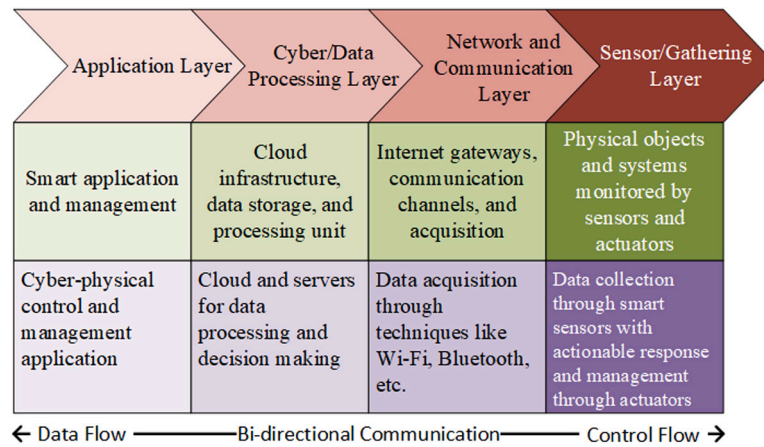


Fig. 9. Layers in IoT Infrastructure.

(CV), lasers, scanners, GPS systems, and other data collection technologies. The ‘internet’-oriented facet of the IoT involves communication technologies that facilitate data exchange between smart devices and connect them to the broader internet. This encompasses technologies such as ZigBee, WiFi, Bluetooth, and cellular communications. The third facet of the IoT paradigm, namely the ‘semantic’ or ‘applications’ feature, manifests in a multitude of applications made possible by smart devices (Chen, Sun, Zhu, Zhen, & Chen, 2012; Yun & Yuxin, 2010).

A standard IoT infrastructure is typically divided into four essential layers: the sensing layer, network and communication layer, data processing layer, and application layer (as depicted in Fig. 9). Each of these layers employs various tools and technologies, allowing for effective bidirectional communication and the implementation of necessary changes to ensure optimal system functionality. Fig. 9 provides a visual representation of the typical composition of an IoT infrastructure. In the past ten years, the IoT has garnered significant attention across a multitude of domains, facilitating the interconnected network of embedded devices that find practical use in daily life scenarios. This technological advancement has ushered in automation in diverse fields, including healthcare, household appliances, security, surveillance, agriculture, and transportation. These applications have comparable effectiveness in extending their reach to the energy sector. IoT has seamlessly integrated itself into the contemporary power grid, often denoted as SGs.

Traditionally, the power grid adhered to a one-way flow of power and information. However, the emergence of DERs, coupled with the advancement of intelligent sensing and communication protocols, has already initiated a transformation in the landscape of the conventional grid. The future grid is thought to be a network of smart sensors and actuators that are all connected to each other. These devices will have built-in intelligence that can understand and solve everyday operational problems with little help from humans. In this context, the smart grid has successfully implemented a comprehensive sensing infrastructure, including Advanced Metering Infrastructure (AMI), alongside robust data processing and transmission technologies. It is well poised to enhance planning, maintenance, and operational procedures by endowing the system nodes with the ability to sense and actuate actions where they are most needed. The subsequent sections delve into the diverse array of applications that IoT frameworks bring to the forefront within the realm of energy and power systems.

The integration of renewable energy into power grids can be seamlessly enhanced through the utilization of IoT. Devices, such as smart meters and sensors, play a vital role in enabling real-time data collection on energy consumption, grid performance, and renewable energy generation. This data empowers grid operators to monitor and optimize grid operations, detect anomalies, and manage energy flows efficiently (Ding, Zeng, Hu, & Yang, 2022; Khan, Haque, Kurukuru, & Saad,

2022). IoT also facilitates energy monitoring and management, allowing consumers to gain insights into their energy usage patterns and optimize consumption in accordance with renewable energy integration goals. IoT devices provide continuous monitoring of grid conditions, which helps to maintain grid stability, manage fluctuations caused by intermittent renewables, and ensure reliable integration. Additionally, IoT-enabled asset monitoring and maintenance enable proactive management of renewable energy assets, optimizing performance and reducing downtime. IoT strengthens grid security by monitoring and detecting cybersecurity threats, contributing to the resilience of the grid (Russell, Goubran, Kwamena, & Knoefel, 2018; Wang, Dabbaghjamesh, Kavousi-Fard, & Mehraeen, 2019). Furthermore, IoT supports regulatory compliance by tracking and reporting data, streamlining certification processes, and ensuring transparency in renewable energy generation and consumption (Pawar et al., 2020). IoT’s role extends to energy trading and market integration, enabling decentralized energy transactions and facilitating the integration of distributed renewable energy sources into the grid. IoT-generated data offers valuable insights for policy development, urban planning, and infrastructure expansion, fostering evidence-based decision-making (Sioshansi, Zamani, & Moghaddam, 2022).

4.1. IoT and power generation

The proposal of challenges and restrictions in power generation, in accordance with the growing grid structure, is a crucial aspect of IoT-based uses of digital technology in power systems. Traditionally, the generation and administration of electricity were understood and implemented via localized control systems. System operators had limited capabilities for remote control, and choices were mostly based on localized orders (Olivella-Rosell et al., 2018). The issue at hand presents a constraint in relation to the effectiveness of operations and the use of automation, particularly considering the always changing landscape of power resources and consumption. In contemporary power systems, the integration of intermittent generating sources like solar PV and Wind farms has become more prevalent. However, relying solely on localized control measures might give rise to greater system-level management challenges for the grid (Zohuri & Zohuri, 2018). The introduction of EVs has contributed to the inconsistent generation of electricity, as well as the emergence of unprecedented intermittency and dynamism in the electric load. This phenomenon has significant implications for the planning, operation, and management of the generating side of the electric power system (Moon, Park, Jeong, & Lee, 2018; Un-Noor, Padmanaban, Mihet-Popa, Mollah, & Hossain, 2017). In summary, there is a pressing need for a grid system that can facilitate the simultaneous transmission of electricity and information in both directions.

In the traditional power grid framework, utility providers relied on local reports from grid personnel or customers to detect irregularities such as demand spikes or service disruptions. In the future SG vision, a network of intelligent sensing devices, expected to span the entire grid, can automatically detect these service disruptions. These intelligent devices, including components like smart metering infrastructure, play a pivotal role facilitated by the IoT. These smart components are equipped with digital markers, such as IP addresses, and possess the ability for two-way communication. The IoT acts as a catalyst, enabling real-time networked communication not only between consumer devices via various communication technologies but also between power equipment through these smart devices. This transformative shift fundamentally alters the grid structure, enabling collaboration among equipment and devices at all levels of the grid hierarchy. This co-operation facilitates real-time, high-speed bidirectional data sharing, culminating in an overarching boost in operational efficiency (Hossein Motlagh, Mohammadrezaei, Hunt, & Zakeri, 2020). Beyond the dynamics introduced by RES and EVs, the electricity pricing mechanism is undergoing transformation, incorporating dynamic pricing structures such as Time-of-Use pricing (Hale, Bird, Padmanabhan, & Volpi, 2018). This shift necessitates real-time data collection and transmission, making it a fertile ground for IoT applications. Furthermore, the emergence of small, isolated distributed generation, particularly within the virtual power plant concept, underscores the need for a smart infrastructure empowered by IoT technologies, capable of seamless data collection, transmission, and actuation.

Power generation is extensively exploring IoT applications in the realm of DERs. The integration of renewable energy generators like solar cells, wind turbines, and photovoltaic cells into the power grid is crucial to addressing environmental concerns. However, the intermittent and location-dependent nature of these sources introduces challenges in predicting and maintaining a reliable power supply. To maximize the reliable and efficient utilization of renewables, researchers are investigating diverse designs, configurations, and strategies. For instance, a study presented in Wu, Yang, Han, and Pan (2022) delved into the design of building-integrated solar generation, leveraging the ZigBee wireless sensor network and IoT to explore distributed solar energy. A shadow radiant energy model was established to enhance understanding and utilization.

Addressing the retirement of solar generation units, a unique approach was taken in Muthusamy, Velusamy, Thandavan, Govindasamy, and Savarimuthu (2022), focusing on recycling processes for solar PV modules using IoT in industries. This investigation employed machine learning, employing k-NN and LSTM network-based learning algorithms to manage smart bin levels within an IoT platform. The smart bin classified solar waste in real-time, notifying collection centers through the IoT platform when predefined thresholds were reached. Furthermore, IoT technology seamlessly connects and integrates renewable energy sources into the grid. Real-time weather information gathered by IoT sensors aids in forecasting energy availability. An IoT-enhanced smart grid system utilizes Kalman Filter-based state estimation and discrete-time linear quadratic regulation techniques to control DER states (Rana & Li, 2015). Additionally, a Web of Things-based SG architecture is proposed to remotely monitor and control renewable energy sources (Mohanty, Panda, & Pattnaik, 2014).

4.2. IoT applications in grid maintenance and monitoring

Effective grid monitoring and maintenance assume critical importance within a highly dynamic landscape of generation and demand. Isolated MGs, where conventional inspection methods may not be feasible, further magnify this significance. As a result, a multitude of IoT applications have emerged, addressing various aspects of modern grid necessities.

A prime example is solar PV maintenance, which has become a focal point of research. Researchers have harnessed the IoT framework in

various ways to oversee and uphold solar PV modules. IoT networks meticulously track key factors such as dust accumulation, shading effects, and power parameters. This real-time, remote, and intelligent monitoring affords a swift and targeted response. Furthermore, the IoT indirectly facilitates the gathering of critical data for advanced predictive modeling, which is necessary considering the grid's inherent instability stemming from intermittent resources and varying demand patterns.

A concrete instance from the literature, as presented in Rao et al. (2018), demonstrates the implementation of a smart solar panel cleaning system utilizing IoT technology. This innovative scheme boasts the ability to detect dust accumulation and assess cleaning efficacy. By executing routine cleaning tasks, the system bolsters maintenance efficiency while concurrently enhancing the overall effectiveness of solar PV panels. In Mokhtar et al. (2020) a smart temperature monitoring system was developed for a Parabolic Solar Thermal Collector (PSTC) integrated with Membrane Distillation (MD) system. The primary goal of the established system was to monitor, record and control the temperature of the salt solution in PSTC. Further, the energy performance was analyzed using the data collected. In the same vein, IoT-aided solutions are being explored in Wind farms. Wind farms are typically offshore infrastructures, and hence the need for remote monitoring and maintenance becomes a chief operational concern. Further, data transmission to central databases without security breaches is also one of the key issues being investigated in the research. In Zhao, Zhou, Matsuo, Korkua, and Lee (2019), the researchers investigated FPGA-CPU-based holistic condition monitoring and SSCI monitoring functions. The technology can record data for post-event analysis as well. Similarly, Sujatha, Deepalakshmi, and Cao (2018) proposed a control scheme for the windmills which convert wind energy to electrical energy. The scheme is constituted by a doubly fed induction generator (DFIG). The IoT-based framework used fuzzy logic to acquire features that involved decision-making logic. Fuzzy logic control is used to introduce human-like flexible decision-making and prove to have intelligent speed control. Further, to enable direct current voltage and to guarantee a pure sine wave for the current on the grid side, a Grid Side Converter (GSC) is used, which is controlled using FLC.

Besides, Noor-A-Rahim, Khyam, Li, and Pesch (2019) explored a state monitoring communications framework within an IoT-based wireless paradigm. A unique fusion algorithm was proposed to combine and process the data gathered from multiple sensors of the perception layer of the IoT. The results obtained from the data were then utilized to predict the state of wind turbines. In terms of fault detection and predictive maintenance, De Sousa et al. (2019) proposed an intelligent incipient faults detection system to be used in wind turbine electric generators to improve maintenance routines. The proposed method employed advanced statistical techniques, namely higher-order statistics (HOS) and the Fourier transform, along with the use of a structural co-occurrence matrix (SCM), for the extraction of features from vibration signals. The effectiveness of several classifiers, including KNN, MLP, OPF, and SVM, were evaluated in terms of their ability to predict the subsequent position of the wind turbine (WT). The accuracy of the classifiers was found to be close to 99%. Similarly, Qian, Zhang, Tian, Si, and Li (2019) novel cloud computing-based condition monitoring system for wind turbines (WT) was proposed in this study. The usefulness of the proposed method was validated using SCADA data obtained from a wind farm. Further, in Rebouças Filho et al. (2019) the possibility of utilizing IoT for the accurate estimation of electricity generation from wind turbines (WT) was explored. Two approaches were proposed for this purpose: (1) Wind Turbine Power Curve Regression and (2) Time Series Data-based Wind Speed Forecast. The results of the analysis showed that the two proposed approaches could be used individually for optimized maintenance planning during periods of low wind speed and to tune the wind turbine controller.

4.3. IoT in microgrid management and control

A MG represents a grouping of small-scale energy generation resources, including elements like photovoltaic panels, wind turbines, and energy storage facilities. These resources cater to local energy needs and have the potential to even distribute surplus energy to the primary grid (Cao, Chen, Qing, Zhang, & Sun, 2019). Due to the limited oversight and control of micro-sources by the main grid, the energy management within a MG operates independently (Moazzami, Ghanbari, Shahinzadeh, Moradi, & Gharehpetian, 2018). To enhance the main grid's ability to monitor and control MG components, elevate power quality, establish a stable interconnection between generation and storage units, thereby boosting the utilization of renewable energy sources and supporting real-time power quality monitoring, the incorporation of the IoT is indispensable (Alavi, Javadipour, & Mehran, 2019). The IoT infrastructure also facilitates the sharing of data from internal MG components with the main grid operator, bolstering their capacity to ensure security in MG interconnections (Hua et al., 2021). Moreover, harnessing intelligent controllers rooted in deep reinforcement learning (DRL) can optimize MG performance. In a simulated environment, DRL-based controllers demonstrated a remarkable reduction of overall generation costs, outperforming conventional methods by 7.1% and 37% (Al-Saadi, Al-Greer, & Short, 2023). On another front, demand-side energy management, often referred to as power demand management, involves modifying consumers' energy usage patterns in response to fluctuating electricity prices set by utility companies (Jaradat, Jarrah, Jararweh, Al-Ayyoub, & Bousselham, 2014). IoT devices gather energy consumption data from diverse household appliances, transmitting this information to control units within the residence. The smart grid control unit then orchestrates the energy consumption schedule of these appliances based on user preferences, ultimately reducing individual electricity bills (Yu, Zou, & Wang, 2016).

4.4. IoT electric vehicles management

EVs have emerged as a promising solution for eco-friendly transportation, notably due to their capacity to substantially reduce carbon dioxide emissions. This provides a distinctive opportunity for the synergy between SG systems and the IoT to seamlessly integrate with EV charging infrastructure, thereby cultivating a transportation network that is both environmentally sustainable and operationally efficient (Balijepalli, Pradhan, Khaparde, & Shereef, 2011). The typical architecture of an EV charging system encompasses a power supply subsystem, charging equipment, and a monitoring subsystem. The power supply subsystem administers and delivers electricity, while the charging equipment facilitates EV charging and discharging via AC and DC chargers. Households typically use AC chargers for slower charging, while public charging stations use DC chargers for faster charging. Both charger types include billing functionalities. The monitoring subsystem supervises the charging process, ensuring its security and enabling real-time oversight. IoT technology plays a pivotal role in this monitoring subsystem, providing an information management framework that harmoniously integrates the diverse components of the charging system. This integration empowers the power supply and monitoring subsystems to transmit data to the information management system, which subsequently relays this information to the control center, facilitating necessary actions. Numerous EV charging systems are leveraging IoT-based technological frameworks. For example, as outlined in Bhaskar, Prasanth, and Saranya (2022), a secure energy management system founded on blockchain technology and IoT connectivity was developed. This system communicates charging levels, travel distances, and vehicle locations through an IoT network to an information processing hub, which employs a predictive classifier to suggest charging tariffs. Furthermore, a forecasting application was introduced by Savari, Krishnasamy, Sathik, Ali, and Aleem (2020), functioning in real-time on a server. This application serves two main

purposes: firstly, it manages schedules to mitigate wait times, and secondly, it provides cost-effective and swift charging recommendations for EVs while ensuring user privacy and circumventing third-party interventions.

To sum up, EVs herald a promising era of environmentally conscious transportation, and the incorporation of IoT technology holds immense potential in augmenting the efficiency and efficacy of EV charging systems. By amalgamating IoT technology with EV charging infrastructure, SG systems can achieve superior integration and optimization, culminating in a more sustainable and effective transportation network. A summarized overview of IoT applications is outlined in Table 5.

5. Case study of AI-enabled, blockchain-based EV integration (Wang et al., 2020)

This case study presents AI-Enabled Blockchain-based Electric Vehicle Integration System (AEBIS), a cooperative platform designed for energy monitoring on a distributed energy platform. The EV fleet plays a dual role within the VPP architecture, serving as both a client and a supplier of electrical power, as depicted in Figure A. This framework employs advanced ANN and FL algorithms to accurately forecast the charging requirements of EVs.

Fig. 10, illustrates the whole structure. The traditional integrator is substituted with a distributed FL-based virtual power-integrator. The utility's network acquires and preserves electric power, from all sources, and subsequently distributes it to customers. Once the green energy offered is no longer enough, the energy firm will notify the electrical grid to supplement the energy delivery using conventional resources. In addition to outlets for charging, EV teams offer a viable method for supplying power back to the electrical network. The suggested integrator intends to optimally use electricity while offering reliable connectivity by interacting with the electrical grid, green power sources, along with the transportation network. The distributed VPP integrator is implemented using a blockchain architecture. This study primarily addresses two fundamental problems: (1) the distributed VPP accumulator communicates with EV teams to execute FL modeling for charging cells recharge forecast. (2) A blockchain system is introduced to substitute the traditional VPP integrator. Within Fig. 10, the thick arcs lines represent the transmission of information on the blockchain. The dotted arcs lines represent the remaining networking actions. An AI-based system utilizing programmable architecture is employed to calculate the anticipated energy usage for every EV on the customer-side for a subsequent time frame. Thus, by considering the present available capacity of an EV as well as the electrical grid's demand, one can determine the precise quantity of energy that the EV must provide. Inadequate electricity on the utility's grid may be determined by the exchange of data among the accumulator and customers, which helps to establish the quantity of energy delivery.

5.1. Assessment approach

The energy utilization forecast information set includes the climate, location, as well as customer data elements. The authors of this study gathered meteorological data spanning the beginning of 2020 to June 2020 in the regions of Japan. The automobile booking had a designated beginning points, and the allowed period of usage bracket from 0 to

Table 5

Classified summary of IoT applications in energy and power systems.

Application classification	Core concept	Challenge area	Year	Ref.
Monitoring and Maintenance	A smart solar panel cleaning system using the IoT technology. The proposed scheme was capable of monitoring the dust detection and cleaning capability. The system has the ability to perform routine cleanup capacity, enhancing maintenance efficiency as well as total efficiency of the solar PV panels.	PV farm monitoring	2017	Rao et al. (2018)
	A smart temperature monitoring system was developed for a Parabolic Solar Thermal Collector (PSTC) integrated with Membrane Distillation (MD) system. The primary goal of the established system was to monitor, record and control the temperature of the salt solution in PSTC.	Temperature monitoring of solar PV	2018	Mokhtar et al. (2020)
	FPGA-CPU based holistic condition monitoring and SSCI monitoring functions. The technology is able to record data for post-event analysis as well.	Holistic condition monitoring	2019	Zhao et al. (2019)
	A control scheme for the wind mills which convert wind energy to electrical energy. The scheme is constituted by doubly fed induction generator(DFIG). The IoT based framework used fuzzy logic to acquire features involving decision making logic.	Advanced electro-mechanical maintenance	2018	Sujatha et al. (2018)
	A state monitoring communications framework within IoT-based wireless paradigm. A unique fusion algorithm was proposed to combine and process the data gathered from multiple sensors of the perception layer of the IoT.	Holistic condition monitoring	2019	Noor-A-Rahim et al. (2019)
Power Generation and RES Integration	An intelligent incipient faults detection system to be used in wind turbine electric generators to improve maintenance routines.	Fault detection in electro-mechanical systems	2019	De Sousa et al. (2019)
	A cloud computing-based condition monitoring system for wind turbines (WT) was proposed in this study. The usefulness of the proposed method was validated using SCADA data obtained from a wind farm. A gearbox condition monitoring model, which considered both gearbox oil temperature and bearing temperature, was established.	Condition monitoring of wind farms	2019	Qian et al. (2019)
	Usage of ZigBee wireless sensor network and IoT in distributed solar energy was investigated and a shadow radiant energy model was established.	Building integrated solar PV generation	2022	Wu et al. (2022)
	Using k-NN and LSTM network based learning algorithm to update and notify the level of smart bin in an IoT platform. The smart bin also classifies the solar waste in real time with the help of industrial internet of things	Solar PV recycling	2022	Muthusamy et al. (2022)
	Kalman Filter-based state estimation and discrete-time linear quadratic regulation method to control the states of DERs	Distributed energy resource	2015	Rana and Li (2015)
Microgrid	A web of things-based SG architecture is also proposed to remotely monitor and control renewable energy sources.	Renewable energy resource monitoring	2014	Mohanty et al. (2014)
	A new method for estimating the state of a MG using a long-range wide-area network (LoRAWAN) protocol has been suggested. This involves using an agent structure based on an event-triggered Kalman filter. A setup has been created to gather large amounts of data from smart meters, which will enable the MG to achieve an optimal state estimation	State-estimation of MG	2019	Alavi et al. (2019)

24 h. We assessed an individual's age bracket based on the criteria specified for obtaining a Class 2 certification. The typical day's energy usage was assessed using the provided input characteristics along with the estimation procedure.

The authors examined the situation in which the information of every customer is distinct that follows the same distribution. We distributed the full information across a total of three customers, with every portion consisting of a thousand data points that match an identical distribution. Nevertheless, in the majority of real-world scenarios, the regional information for every EV junction is often non-independent and identically distributed. This is because the information is acquired at various times or by various individuals. Consequently, we examined the influence of non-independent and identically distributed statistics on the result. An intriguing scenario arises once every EV is scheduled at distinct time intervals throughout the day, such as early dawn, lunchtime, evening, as well as the end of the day Every FL train session had fifty repetitions of the scenario. They employed the R2 Score as a metric to evaluate the system's functionality.

5.2. Assessment results

In this section, we shall analyze the evaluation findings of the case study. The result of the first test, in which every customer's information is autonomous and uniformly distributed, is displayed in [Fig. 11](#). The FL method demonstrates a median R2 Score outcome of 0.922, that is lower compare to the traditional design efficiency of 0.938. The precision loss can be attributed to the somewhat skewed distribution of the information. Tests of the FL system on non-independent and identically distributed information distributions are depicted in [Figs. 12 and 13](#). Based on this assessment, the authors note that the outcome of one customer declines considerably as opposed to the traditional system. This decline is attributed to the presence of highly tilted, non-independent, and identically distributed information. Nevertheless, the framework based on FL has demonstrated its resilience in both scenarios. In the present EV pooling group, precise energy prediction enables users to anticipate the remainder of power in EVs ahead of time. As a result, the administration of reservations gets better.

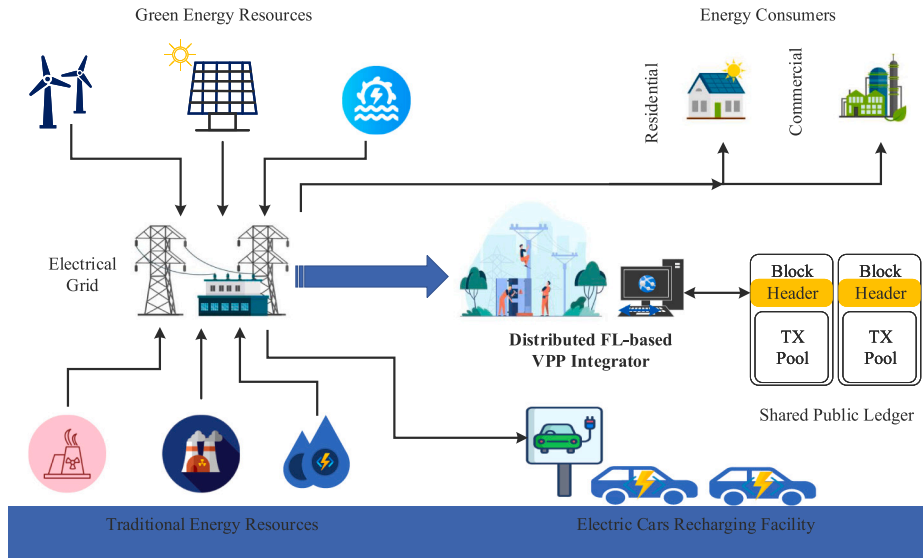


Fig. 10. AI-enabled blockchain-based EV cooperative Design.

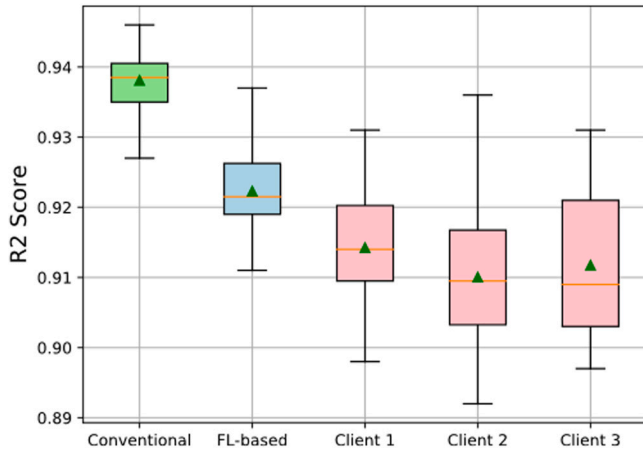


Fig. 11. Examining the traditional approach in contrast to the individual learning approach (Wang et al., 2020).

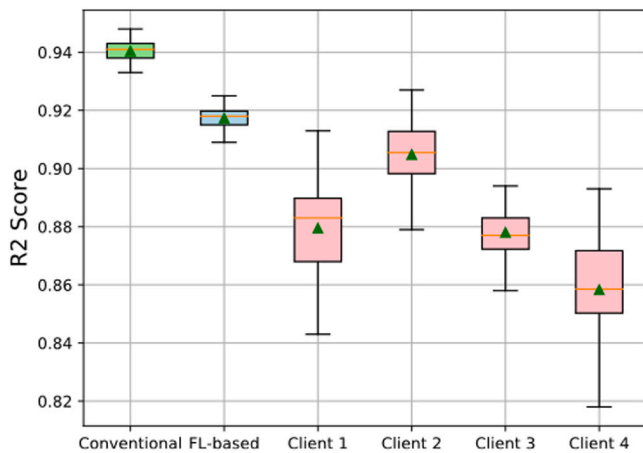


Fig. 12. Examining the traditional approach in contrast to the individual learning approach (Wang et al., 2020).

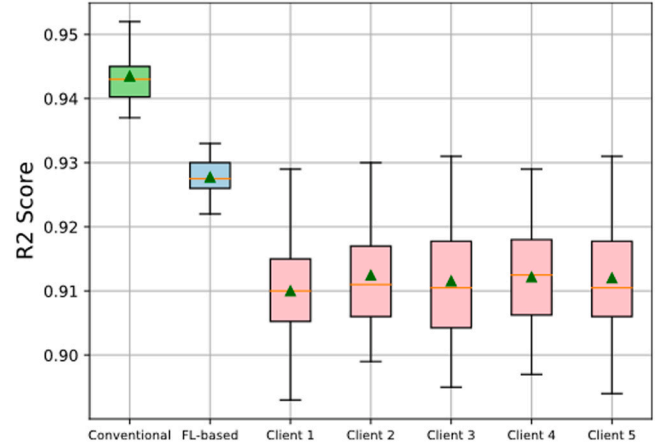


Fig. 13. Examining the traditional approach in contrast to the individual learning approach (An age range of twenty-five to ninety-nine is connected with every customer in the information set) (Wang et al., 2020).

5.3. Summary of case study

The FL paradigm effectively resolves the challenges arising from non-independent and identically dispersed information as well as fractional equipment engagement. In addition, the process of sharing prototypes without disclosing the initial information set is seen as a means of safeguarding customer information confidentiality to a certain degree. The findings suggest that proposed FL-based technology can offer a precise forecast for EV electrical usage, and this is essential for both future energy delivery and EV allocation. Nevertheless, there remains some possible risks that remain owing to malevolent customers and hacking.

6. Discussion and analysis

The full review above shows how digitalization can modernize, automate, and improve grid intelligence. By enabling real-time information assessment, improving energy consumption, along with enabling safe communications and information management, the addition of AI, blockchain and IoT computing improves energy and control applications' efficiency and environmental sustainability. Blockchain

promises reliable and open transactions as well as data management hence enhancing system dependability and reducing power loss. On the other hand, AI algorithms can analyze IoT-generated information in order to enhance energy utilization and to predict user demand. When studied individually, these applications have their own challenges. These concerns hinder energy and power system adoption. The relevant technology has several issues. The wide-ranging implementation of AI, blockchain, as well as IoT approaches in the energy market is disturbed by numerous common constraints which includes fears concerning information security, extensive implementation expenditures, challenges associated with collaboration amongst various approaches and frameworks, lack of standardized procedures, as well as the necessity for talented staffs to supervise and tolerate these approaches. Besides, the application of these technologies is delayed by regulatory obstacles as well as the unwillingness to embrace change in conventional areas. A significant number of challenges are connected to the incorporation of AI, IoTs, and blockchain technology into the control of the market as well as the supply and demand of power.

1. **Data Security and Confidentiality:** The rising use of IoT tools and the reliance on AI-based algorithms for data assessment have both contributed to the emergence of data concerns that compromise data security and confidentiality. The technology known as blockchain has the potential to enhance the safety of the transactions that include information. Furthermore, safeguarding the privacy of Internet of Things devices and the dependability of AI systems continues to be a difficult undertaking.
2. **Interoperability:** The formulation of policies for interoperability is frequently required in order to ensure that the integration of various devices and networks is successful. These policies are designed to ensure that information is transmitted and interactions are carried out without any hiccups. There is a possibility that establishing interoperability between blockchain, IoTs, and AI technologies may present complex obstacles and result in considerable expenditures.
3. **Scalability:** When there is a growth in both the number of IoTs devices and the complexity of ML algorithms, stability becomes a significant issue. In order to properly govern the market and keep the supply of power stable, it is essential to ensure that the network is able to accommodate the growing volume of data and transactions.
4. **Energy Utilization:** It is possible that the surge in energy use is due to the increased processing needs of AI programmes and blockchain technology. It is necessary to find a solution to the problem of addressing the power needs of such devices in order to achieve cost-effectiveness in grid management.
5. **Fairness and Flexibility in the Market:** The utilization of blockchain computing has the potential to enhance the liability and fairness of energy markets by establishing a reliable and permanent record of transactions. This might be accomplished through the utilization of blockchain technology. Despite this, the challenge that still has to be addressed is making sure that the tools are utilized in a manner that encourages transparency and equity.

Coordination among several stakeholders is essentially important to face these challenges. The establishment of standards and ideal procedures for the incorporation of AI, IoT, and blockchain-based technologies can successfully report possible dangers and promise the seamless implementation of these developments in the areas of power grid reliability as well as market administration.

The other challenges that AI approaches now encounter is the issue of data consistency, which must be approached from a holistic perspective. The maintenance of performance and dependability in AI-based digitization within electrical grids demands the task of ensuring information precision. With the intention of achieving this, it is required to employ accurate monitoring of data quality procedures. The procedures

involve information collection, authentication, as well as normalization procedures. The constant data inspection plays a fundamental part in the detection and resolution of faults, consequently verifying the capable and dependable operation of AI-based systems in electrical systems. The topic at hand is tricky since certain approaches are not replicable, which prevents their wider adoption. However, in areas where data gathering and processing are well developed, AI is demonstrating encouraging outcomes. It is important to determine an effective plan for policy formulation and power planning for transitions to successfully tackle the problem surrounded by the electrical business. This specific method presents an organized methodology for successfully delivering the complexities linked with advancing the grid as well as the shift concerning clean energy sources. The method can accomplish alignment of investments in addition to activities concerning coordinated objectives by describing a definite enduring vision, engaging applicants, combining planning programs, as well as modifying the legal framework. The successful integration of IoT and blockchain computing into electricity and utility systems severely depends on the cruciality of information security. The data security acquired by IoT devices is guaranteed, the precision of information is preserved by the impermeable maintaining records of blockchain, as well as the identification and accessibility of both devices and individuals in the system are managed. Furthermore, the capabilities of AI are constrained by existing computer power. It is evident that computational capacity represents a significant obstacle in the progression towards achieving a highly sophisticated AI system. Hence, it is imperative that policy deliberations encompass more than the mere development of AI techniques and algorithms in isolation. The integration of computational power within AI programs at both national and international levels is of utmost importance. Both IoT and blockchain applications possess inherent constraints. However, the primary constraint they face is the progress of AI. AI comprising algorithmic learning techniques can enhance the information safety in IoT and blockchain by recognizing irregularities, predicting repairing requirements, inspecting user actions for possible confidentiality hazards, along with enabling encrypted information transmission. By inspecting previous information patterns and providing perceptions for improving operations and decision-making procedures, these tools improve prospective and prescriptive information-based operations. In order to operate efficiently inside large-scale systems, such as the electric power system, data churning technologies necessitate more data processing capability as well as predictive and prescriptive analysis of the generated data. The resolution of this issue cannot be achieved only by conventional data processing approaches. As a result, several investigations pertaining to the utilization of these technologies in domains such as P2P trade and smart grids have been constrained to smaller datasets and rudimentary calculations.

AI enables the transformation involving a decentralized grid which is decarbonized and computationally improved by developing the energy administration, supporting system reliability, as well as advancing the addition of energy efficient sources. Also, it facilitates an effective control of supply as well as improves the control of resources, thus contributing to the development of a robust and environmentally friendly power network. Additionally, AI advances cybersecurity by preserving the electrical grid from cyber-attacks with ensuring the integrity of critical systems. Additionally, the integration of AI, ICT, and power facilities is presumed to necessarily change the manner energy is generated, supplied, and utilized in future. AI algorithms have the facility to improve the production and delivery of electricity which results in enhanced economy and reduced prices. ICT aids continuous tracking and regulation of power infrastructure, therefore improving the operation of electricity flow. AI-based digitization significantly improves the reliability of power as well as energy systems by means of the combination of digitally enabled trading techniques. These methods deliver improved electricity market, successfully achieving the balance among energy availability and utilization in actual time. Hence, the grid becomes stable and reliable, especially when green energy sources are integrated. Individuals are authorized to be involved in the energy sector via cyber trading promising the acceptance of useful costs practices.

7. Conclusion

The aim of this study was to investigate various applications in the field of energy and power systems, with a specific emphasis on three significant digital technologies: AI, blockchain, and IoT. The examination revealed a significant increase in research efforts in this field. Our analysis provided further evidence supporting the claim that common obstacles hinder the widespread adoption of these technologies in various electrical markets. Although each technology demonstrates significant potential for enabling the transition toward a decentralized, decarbonized, and digitally upgraded grid, certain shared limitations need to be considered. The observed progress in the field of AI presents significant benefits; however, the ongoing issue of ensuring data consistency can operate as a constraining element. Furthermore, the complex domains of data security and the predictive and prescriptive data processes inherent in the IoT and blockchain necessitate the involvement of AI and ML methodologies in their resolution. This highlights the need for a more holistic perspective on digitalizing the electrical industry. The progressive integration of these technologies, while beneficial, may present a scaling issue for the electricity industry. This method can cause grid instability and market management concerns. Based on these factors, we recommend a complete policy development and grid-transition-planning strategy. Implementing a smart grid without integrating strong computing capabilities, advanced AI methodologies, or digital market trading practices can result in less-than-ideal outcomes. Both commercial and academic research endeavors need to be in accordance with this comprehensive viewpoint, thereby ensuring a more all-encompassing and efficient advancement toward the future of energy and power systems.

CRedit authorship contribution statement

Muhammad Khalid: Conceptualization, Data curation, Formal analysis, Methodology, Validation, Visualization, Writing – original draft, Writing – review & editing.

Declaration of competing interest

The authors state that they have no financial or personal conflicts that might have influenced their research.

Data availability

No data was used for the research described in the article.

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